



CHECKLIST OF TENDER DOCUMENTS

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1	GeM Tender Notice	02 Pages
2	Agree Terms and Conditions	03 Pages
3	Scope of Work	03 Pages
4	Technical Specifications (Annexure-I, II & III)	33 Pages
5	Eligibility Criteria	01 Pages
6	Bill of Quantities	02 Pages
7	Bid Security Declaration	01 Pages
8	Declaration by Vendors along with Bid (Mandatory) on their Letter Head in Case the value of Procurement Is more than 5 Lakhs	01 Pages
9	General Terms and Conditions	08 Pages
10	Make In India Order	09 Pages
11	Commercial Offer (Un-priced)	01 Pages
12	Price Breakup BOQ (to be submitted in Cover-II/Price Bid)	01 Pages
13	Price Bid Format	01 Pages
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MISHRA DHATU NIGAM LIMITED
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NIT DOCUMENT

GEM-TENDER NOTICE

Tenders (online bid) are invited through GeM portal from the eligible bidders from any agency authorized by Controller of Certifying Authority (CCA), Govt of India.

Sl. No	MATERIAL Description & Specification	Estimated Requirement	
		QTY	Units
1.	Design and Supply of materials for Fire Fighting Systems for Shed-4 & Admin Building at Rohtak plant as per (Annexure-I,II,III)	01	SET
2.	Installation, Erection, Commissioning & Testing of the above systems as per tender	01	LUMPSUM

Specification	As per Annexure-I, II, III
Delivery Required By	Material Supply within 60 days from the date of receipt of PO & Installation, Erection and Commissioning within 60 days from the date of site clearance.
Terms of Payment	FOR Supply: 90% payment along with 100% taxes, within 30 days from the date of receipt and acceptance of the material at MIDHANI. For Service: 90% payment along with 100% taxes, within 30 days from the date of erection & commissioning on Indenter's certification. Balance 10% payment of both supply and service shall be done after acceptance of the material at our end, successful commissioning and against submission of Performance Bank Guarantee for 10% value of the order valid till Guarantee / Warranty Period.
Validity of the offer	The offer should be valid for a minimum period of 180 Days from the date of opening of tender.
Acceptance Criteria	As per Specification and Scope of Work.
EARNEST MONEY DEPOSIT (EMD AMOUNT):-NIL The tenderer/bidder shall furnish EMD, as mentioned below at the time of submission of bid. For complete details, refer GENERAL TERMS AND CONDITIONS document clause No. 17 EMD Exemption: Procurement under this bid is reserved for purchase from Micro and Small Enterprises, whose credentials are validated online through Udyog Aadhaar for that	

product/service category. If the bidder wants to avail the reservation benefit, the bidder must be the manufacturer of the offered product in case of bid for supply of goods. Traders are excluded from the purview of Public Procurement Policy for Micro and Small Enterprises. Relevant documentary evidence in this regard shall be uploaded along with the bid in respect of the offered product or service.

SECURITY DEPOSIT (SD): The successful tenderer/Bidder shall be required to **furnish SD amount of 10% of purchase order value** within 21 days from the date of Purchase Order.

For complete details, refer GENERAL TERMS AND CONDITIONS document clause No. 18

SD Exemption: All Central PSU and Govt. departments may be exempt from submission of SD in all tenders.

Note: If you are submitting Bank Guarantee (BG) in Place of SD, BG shall have a claim period of one year from the date of its expiry. In case, SD is not submitted within 21 days from the date of PO, interest@12% shall be levied for the period of delay beyond 21 days.

CONTACT DETAILS

For any Technical Clarifications:

Mr Mallareddy Marri, Ph:040-2418 4525

For Commercial and Tender Related Clarifications:

Shri Raj Kumar (DGM-Purchase) Ph:040-2418 4578

E-mail: rajkumar@midhani-india.in

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E-mail: diptimoy.mohanty@midhani-india.in

Contact-7013952395

For Make In India & Vendor Registration Clarifications:

Shri Raj Kumar (DGM-Purchase) Ph:040-2418 4578

Email: rajkumar@midhani-india.in

Note:

1. Please refer for complete specification, tender terms & conditions and other additional documents under work item section.
2. MIDHANI Reserves the right to conduct discovery through OLSB or OLSB Submission, followed by RAP
3. Payment shall be released only after filing of GSTR1 and GSTR3B including availability of invoices in GSTR2A of MIDHANI

AGREED TERMS & CONDITIONS (ATC)			
TENDER TITLE		Design, Supply, Erection, Commissioning and Testing of Fire Fighting Systems for 4th Shed & Admin building at Rohtak plant per Annexure-I, II & III	
Bidder Name:			
S.NO.	Tender Requirement	Vendor's confirmation (ARGREE /DISAGREE)	REMARKS in Max 500 Characters
*1	Supply the material as per MIDHANI Tender Quantity and Specification without any deviation.	Select	
*2	Specify Agree/Disagree of GEM Tender Notice Inviting Tender (NIT) All the Terms and Conditions.	Select	
*3	Basis of quoted prices. FOR (FREE ON RAIL/ROAD) MIDHANI, ROHTAK-124001	Select	
*4	Payment Terms as per NIT	Select	
*5	Validity of Quote- 180 days	Select	

*6	Delivery Schedule as per NIT	Select	
*7	Bidder are informed to comply the "Make In India" (MII) General Terms & Conditions (GTC) Clause No.19. As per GTC 19(f) the Class-I Local Supplier and Class-II Local Supplier has to provide the following details in bidder company letter head with the authorized sign and stamp of the organization. (i) Tender Reference No_____Date_____ (ii) Percentage of Local Content.-----	Select	
*8	Bidder shall furnish Earnest Money Deposit (EMD) as per NIT	Select	
*9	Integrity Pact along with Undertaking form has to be sign & stamped by the respective bidder by mentioning the tender reference number & date details and the same document has to be uploaded in the e-tenderng portal.	Select	
*10	Bidder has to comply all the General Terms & Conditions (GTC) of the document.	Select	
*11	Unconditional acceptance of all commercial terms & Conditions	Select	
*12	SPECIFY OFFER MAKE and PRODUCT SPECIFICATION		

*13	Indicate PO Address (to be released), GST NO. No. & HSN & SAC Code (Item Wise)	
*14	Indicate <u>Taxes and Duties</u> applicable with rate percentage.	
*15	Any Other Terms / Deviations (If Any).	
*Confirm your acceptance of our terms (S.NO. 1 to 10) by selecting AGREE option , if DISAGREE enter your remarks in respective cell.		
** For Sl no 11 to 15 enter only remarks in respective cell.		
Tenderer's offer must conform in all respects with the applicable specifications and terms and conditions(T&C) of the tender. In case of, deviation if any, from the tender specifications or terms and conditions must be clearly and explicitly stated. Technical deviations and Commercial deviations shall be furnished at SNO.15 (Above) Only along with reasons for taking such deviations. Deviation(s) mentioned elsewhere shall not be accepted. T&C agreed at ATC are final.		

MISHRA DHATU NIGAM LIMITED: ROHTAK

FIRE FIGHTING SYSTEMS FOR 4th SHED & ADMIN BUILDING

SCOPE OF WORK

Supply, installation, testing, commissioning of Fire Fighting Systems for 4th Shed & Admin building 1st and 2nd floor as per the technical specification.

- **Location:** Material should be installed at Midhani at Rohtak Plant, Haryana State
- For all items mentioned in CPR- the order will be placed on overall Lowest (L1) bidder only.
- This system to be paneled/connected to the existing Fire Fighting systems available in other sheds and also connected to centralized Fire control room.
- Technical Specifications is as per the Annexure-I, II & III

I. COMPLETION PERIOD

Supply the material within 60 days from the date of P.O & Installation and Commissioning within 60 days from the date of site clearance.

II. INSTALLATION, COMMISSIONING & TRAINING

The installation, testing and commissioning of the equipment/fire protection system shall be carried out by competent engineers/technicians of the Tenderer at the work site. After commissioning, the party's engineer / technician shall impart necessary training to Owner's personnel in operating and maintaining the installed fire protection system. No separate charge shall be payable by the Owner for the purpose.

III. QUANTITY VARIATION

The quantity as mentioned in the Schedule of Work/ Price Bid is indicative. The selected bidder/Contractor shall however ascertain the exact quantity required at site and supply and install accordingly. As the work progresses, it is possible that there will be quantity variations to any extent & omission of items. Under all such circumstances, the rates should be firm. The quantity of tolerance shall be +/-10%.

IV. PERFORMANCE GUARANTEE:

1 Performance Guarantee:

- a) The Contractor shall guarantee that the equipment and workmanship of work done and any fittings designed / manufactured / supplied by him are as specified in the tender

schedule and wherever there is nothing specifically mentioned shall correspond to the best available grade and quality as required for the application.

b) The Contractor shall also guarantee that the work done and any fittings designed, manufactured, supplied, erected shall be as per prevailing relevant standard, codes and statutory practices / stipulations.

2. Warranty:

The Contractor will repair and/or replace all defective parts, components / fittings, accessories etc. which shall be notified to them in writing within the Defect Liability Period Promptly at free of cost. The Contractor will provide similar warranty on the parts, components, fittings, accessories etc. repaired and/ or replaced.

3. General Notes for Bidders:

(Bidders should confirm each & every point clearly. Deviations, if any, should be highlighted in the quotation.)

1.0 Bidders other than the Original Equipment Manufacturer (OEM), must enclose proper authorization certificate.

2.0 The bidder to provide 3 (three) sets of technical literature for operation, maintenance & repair indicating all components with schematic drawings, part number etc. along with offer.

2.1 Three (3) sets of instruction cum operation manual, recommended parts list etc are to be supplied along with the supply in the event of order. Bidder has to confirm the same in the technical bid.

4. Warranty / Guarantee:

The Systems must be under guarantee / warrantee by the bidder for a minimum period of not less than 18 months from the date of dispatch or 12 months from the date of successful commissioning of the complete unit at site, whichever is earlier. Defective goods / materials or parts notified by Midhani to the Seller shall be replaced immediately by the Seller on F.O.R destination basis including payment of all taxes and duties at Seller's expense. This guarantee shall survive and hold good notwithstanding inspection, payment for and acceptance of the goods.

5. Pre-dispatch Inspection & Testing Charges:

The material shall be inspected by Midhani deputed engineer at manufacturers' works / factory prior to dispatch. However, such inspection will not relieve the supplier of his responsibility to ensure that the equipment supplied conforms to the correct specifications and is free from manufacturing and all other defects. Test certificates and controllers shall have to be forwarded along with delivery of material.

6. Installation & Commissioning:

Installation & Commissioning charges must be quoted separately on lumpsum basis which shall be considered for evaluation of the offers. While quoting Installation & commissioning charges, bidders should take into account all charges including to & fro fare,

boarding / lodging, local transport at Rohtak and other daily expenses of the commissioning personnel.

7. Care of building:

Care shall be taken while handling and installing the equipment to avoid damage to the building. On completion of the installation, the contractor shall arrange to repair all damages to the building caused during installation so as to bring to the original condition.

8. Painting and protection:

Metal work of all equipment to be supplied (except the detectors) shall be given the final coat of paint over the primer after necessary treatment at the works before dispatch. All damages to painting during transport and installation shall be set right or repainted to the entire satisfaction of the Engineer-in-Charge before handing over.

9. PRE-QUALIFICATION CRITERIA:

1. The bidder should have supplied /taken up minimum two installations of similar type of works within 5 years with minimum of Rs. 8.32 lakhs.
2. The parties should give the details of the works taken up with full details of the contact persons along with telephone nos. Supporting documents for successful completion of the work shall be enclosed along with offers.
3. Midhani has the right to disqualify any supplier depending on the supporting documents/ after assessing the performance of systems supplied.
4. Midhani may depute their Engineers to those installations to assess the work done, and the party has to arrange the visit. Expenses for visits if any will be borne by Midhani

ANNEXURE-I

TECHNICAL SPECIFICATION

FOR

FIRE FIGHTING WET RISER

HYDRANT

&

SPRINKLER SYSTEM

TECHNICAL SPECIFICATIONS

1.01 FIRE FIGHTING WORKS:

FIRE PROTECTIONS

1. SCOPE OF WORK

The scope of work covers the supply, installation, testing & commissioning of Fire Fighting Wet Riser Hydrant & Sprinkler system proposed for the Building. It will be the responsibility of the Contractor to get all approval and completion certificate from the Local Fire Department without which the work will not be taken over by the owner. Fee payable to the local bodies for such activities shall also be borne by the owner on production of receipts for money paid and the all other expenses barring the fee will be borne by the contractor.

2. TENDER DRAWINGS

For guidance of the bidder, drawings as listed in Annexure are enclosed with these tender documents. These drawings are broadly indicative of the work to be carried out. The contractor on award of work will furnish detailed stage-wise working drawings as required in advance for approval of Engineer and get the same approved by Local Fire Authority/other statutory bodies. No claim whatsoever shall be admissible on account of changes that may be introduced by the Engineer/ Local Fire Authority.

3. SHOP DRAWINGS

The contractor shall prepare and furnish all shop drawings in quadruplicate at no extra cost for approval by the Engineer before commencing fabrication/ manufacture of the equipment. Such shop drawings shall be based on the Architect drawings and requirements laid down in the specifications and as per site conditions. The manufacture of equipment shall be commenced only after the shop drawings are approved in writing by the Engineer. Such drawings shall be co-ordinate with all disciplines of work.

4. COMPLETION AS BUILT DRAWINGS

On completion of the work and before issuance of certificate of virtual completion, the contractor shall submit to the Engineer. General layout drawings, drawn at approved scale indicating layout of pump house piping and its accessories "As installed". These drawings shall in particular give the following:

- a. General layout of pump house
- b. Panels and other equipment location and sizes etc.
- c. Complete schematic as installed.
- d. Location of Hydrants, Earth pipes, route of earthing conductors etc.
- e. Route of all cables and pipes run along with detail sizes and mode of installation.

5. DOCUMENTS

The contractor shall submit to the Engineer, the following documents on completion of the work and before issuance of virtual completion.

- i. Warranty for equipment installed.
- ii. Test certificates
- iii. History sheets of the equipments

- iv. Catalogues
- v. Operation and maintenance manuals
- vi. List of recommended spares and consumables
- vii. Reconciliation statement
- viii. All approvals and sanctions

6. SANCTION/ APPROVALS FROM STATUTORY AUTHORITIES/ LOCAL FIRE AUTHORITY

The contractor shall be fully responsible and shall carry out following activities:-

- a. Submission of working drawing
- b. Obtaining the approval of drawings
- c. Arranging inspection of site by officials of the Authority
- d. Obtaining the final no objection/ completion certificate after submitting required documents.
- e. Any other statutory approvals required.

7. MANUFACTURING

The responsibility for ensuring the manufacture of the equipment as per the specifications shall be solely that of the contractor. The contractor shall be responsible for selection of materials as per agreed specifications.

8. MAKE OF MATERIALS

Only approved make of material shall be used. The contractor shall get the samples of all the items approved from the consultant or project incharge engineer before commencing the supply.

9. MANUFACTURER INSTRUCTION

Any specific instruction furnished by manufacture covering the points not mentioned in technical specifications of the tender shall be brought to the notice of project incharge engineer in writing for further instructions in this regard at the time of tendering.

10. MATERIAL TESTING

The project incharge engineer shall have full power to get any material of work to be tested by an independent agency at contractor's expense in order to prove the soundness and adequacy.

11. INSPECTION AND TESTING

- a. All equipment shall be inspected and tested as per an agreed Quality Assurance Plan before the same is packed and dispatched from the contractor's works. The contractor shall carry out tests as specified/ directed by engineer.
- b. Contractor shall perform all such tests as may be necessary to meet requirements of Local Authorities, Municipal or other statutory laws/ bye-laws in force. No extra shall be paid for these.
- c. The project incharge engineer may, at his sole discretion, carry out inspection at different stages during manufacturing and final testing after manufacturing.

- d. Approvals or passing of any inspection by the engineer or his authorized representative shall not, however, prejudice the right of the engineer to reject the plan if it does not comply with the specification when erected or give complete satisfaction in service.

12. TRAINING OF DEPARTMENT PERSONNEL

- a. The contractor shall train the owner's personnel to become proficient in operating the equipment installed. Training shall be done before the expiry of the defects liability period (one year after completion & handing over).
- b. The period of training shall be adequate and mutually agreed upon by the engineer and contractor.
- c. The owner's personnel shall also be trained for routine maintenance work and lubrication, overhauling, adjustments, testing, minor repairs and replacement.
- d. Nothing extra shall be paid to the contractor for training owner's personnel.

13. PERFORMANCE GUARANTEE

At the close of the work and before issue of final certificate of virtual completion by the engineer, the contractor shall furnish written guarantee indemnifying the owner against defective materials and workmanship for a period of one year after completion and handing over. The contractor shall hold himself fully responsible for reinstallation or replace free of cost to the owner.

- a. Any defective material or equipment supplied by the contractor.
- b. Any material or equipment supplied by the owner which is proved to be damaged or destroyed as a result of defective workmanship by the contractor.

1.02 PIPING FOR WET RISER SYSTEM

1. SCOPE

This section covers the details of requirement of piping used in wet riser system, including the associated auxiliary equipment.

2. GENERAL

The wet riser system shall remain pressurized at all times during operation, and as such the piping work shall be carried out to withstand the same.

3. PIPES AND FITTINGS

Pipes for Wet Riser system shall be of black steel conforming to IS: 1239 (Heavy Class)

Fittings for black steel pipes shall be malleable iron suitable for welding or tapered screwed threads.

4. JOINTING

Joint for black steel pipes and fittings shall be metal to screw grid up to 50 mm dia and above 65 mm dia welded joints. A small amount of red lead may be used for lubrication and rust prevention in threaded joints. Hold tight will be use for threaded pipes joint.

All the welding shall be radiographic ally tested. Joints between MS pipes, valves and other appurtenances, pumps etc. shall be made with M.S. flanges with appropriate number of bolts. Flanged joints shall be made with 3mm thick insertion rubber gasket.

5. DIA OF FLANGE AND HOLE CONFORMING IS:

Size of pipe	→	80 mm	100 mm	150 mm	200 mm	300 mm
Dia of flange	→	200 mm	220 mm	285 mm	340 mm	445 mm
Dia of bolt	→	16 mm	16 mm	16 mm	16 mm	16 mm
No. of hole	→	4 mm	4 mm	8 mm	8 mm	12 mm

6. PIPE PROTECTION

- All pipes above ground and in exposed locations shall be painted with one coat of red oxide primer and two or more coats of synthetic enamel paint of approved shade.
- Pipes in chase or buried underground shall be painted with two coats of hot bitumen, wrapped with bituminous pypkote or Hessian cloth and finished with one coat of hot bitumen paint.
- Pipe passing through structural members will be provided with M.S. pipes.

7. PIPE SUPPORTS

All pipes shall be adequately supported from ceiling or walls from existing inserts by structural clamps fabricated from M.S. structurals e.g. rods, channels, angles and flats. All clamps shall be painted with one coat of red lead and two coats of black enamel paint. Where inserts are not provided the contractor shall provide anchor fasteners.

Pipe Support Spacing	Horizontal	Vertical
Pipe up to 50 mm	2 Mtr.	3 Mtr.
Pipe 65 – 100 mm	1.75 Mtr.	3 Mtr.
Pipe above 100 mm	1.50 Mtr.	3 Mtr.

8. ORIFICE FLANGES

Contractor shall provide orifice flanges fabricated from 6 mm thick stainless steel plates on the branch lines feeding different zones/ floors so as to allow required flow of water at 3.5 Kg/ sq.cm. pressure. The contractor shall furnish design for these orifice flanges.

9. AIR VESSEL AND AIR RELEASE VALVE

Air vessel on top of each wet riser piping shall be installed before execution for approval fabricated out of at least 8 mm thick steel to withstand the pressure, with dished ends and supporting legs. This shall be of 250 mm dia and 1 m high. This shall be completed with necessary flange connection to the wet riser piping and air release valve with necessary piping to meet the functional requirement of the system. The air vessel shall be ;of continuous welded construction and painted with red Colour. This shall be tested for twice the working pressure.

10. VALVES, GAUGES AND ORIFICE PLATES

Butter-fly or Sluice valves above 50 mm shall be of cast iron body and bronze/ gunmetal seat. They shall conform to type PN 1.6 of IS: 13095,780, valves up to 65mm shall be of gunmetal construction. Valve wheels shall be of right hand type and have an arrowhead engraved or cast thereon the direction for turning open and closing.

Non-return valves shall be of cast iron body and bronze/ gunmetal seat. They shall be swing conform to Class 1 of IS: 5312 and have flanged ends. They shall be swing check type in horizontal runs and lift check type in vertical runs of piping. They shall not be spring-loaded type.

Pressure gauge of suitable range shall be installed on the discharge side of each pump vacuum gauge shall be provided on suction side for pumps with negative suction. The dial size shall be 250 mm. The gauges shall have brass cocks.

Orifice plates shall be of 6mm thick stainless steel to reduce pressure on individual hydrants to operating pressure of 3.5-kg/ sq.cm. Design of the same shall be given by the contractor as per location and pressure condition of each hydrant.

11. EXTERNAL YARD HYDRANTS

External yard hydrants shall be of 'Stand Post' type conforming to IS: 908 and comprise stand post for single or double(as per specified in boq) outlet, duck foot bend, flange riser and single headed brass/ gunmetal or (as per specified in boq)valve conforming type A of IS: 5290.

The stand post column shall be of cast iron, cast in one piece, conforming to grade 20 of IS: 210 or M.S. pipe. The internal diameter at the top shall be at least 80 mm.

The outlet shall be angled towards ground, with instantaneous spring lock type gunmetal female coupling of 63 mm dia. For connecting to hose pipe.

12. INTERNAL HYDRANTS

The internal hydrant outlet shall comprise double-headed double outlet or as per B.O.Q. gunmetal or SS landing valve' conforming to type A of IS: 5290. Separate valves one on each of the two heads shall form part of the landing valve construction.

A brass cap with chain is provided on one head of the outlet which will have an instantaneous pattern female coupling for connection to the hose pipe. The landing valve shall be fitted to a tee connection on the wet riser at the landing.

13. FIRST AID HOSE REEL EQUIPMENT

First aid hose reel equipment shall comprise reel hose guide fixing bracket, hose tubing globe valve, stopcock and nozzle. This shall conform to IS: 884. The hose tubing shall conform to IS: 1532.

The hose tubing shall be 20 mm dia and 36 m long. The gunmetal / brass nozzle and globe valve shall be of 25 mm size.

The fixing brackets shall be of swinging type. Operating instructions shall be engraved on the assembly.

14. HOSE PIPES, BRANCH PIPES AND NOZZLES

Hose pipes:- Hose pipes shall be rubber lined woven jacketed 63 mm in diameter and 15 m long. They shall conform to controlled percolation type comply with IS:8423 or type A (reinforced rubber lined) of IS: 636. The hose shall be sufficiently flexible and capable of being rolled.

Each run of hose pipe shall be complete with necessary coupling at the ends of match with the landing valve or with another run of hose pipe or with Branch pipe. The coupling shall be of instantaneous spring lock type.

Branch pipe: - Branch pipe shall be of copper, gunmetal or aluminum alloy 63 mm dia and be complete with male instantaneous spring lock type coupling for connection to the hose pipe. The branch pipe shall be externally threaded to receive the nozzle.

Nozzle: - The nozzle shall be of copper or gunmetal, 20 mm in internal diameter. The screw threads at the inlet connection shall match with the threading on the branch pipe. The inlet end shall have a hexagonal head to facilitate screwing of the nozzle on to the branch pipe with the nozzle spanner.

End couplings, branch pipes, and nozzles shall conform to IS: 903. Each hydrant point will be provided with two hoses of 15 m each and one gunmetal branch pipe.

15. HOSE CABINET

The hose cabinet to accommodate the hosepipes, branch pipe nozzle and the hydrant outlets shall be fabricated from 1.5 mm thick sheet steel. In case of internal hydrants, this shall accommodate the hose reel equipment also. This shall have lockable, center opening glazed doors.

The scope of work includes provision of masonry or steel frame structure, as specified for installation. The hose cabinet shall be painted red stove enameled.

16. FIRE BRIGADE INLET CONNECTIONS/ DRAW OFF CONNECTION

One set of 2/4 ways collector head Fire Brigade connection shall be provided at underground tank, sprinkler system and individual wet risers as specified.

The inlet to the wet riser sprinkler header shall be with 150 mm dia butterfly or sluice valve and non-return valve. The scope shall include necessary reducers, tees bends and special fittings as required.

It should be provided with M.S. enclosure fabricated from 1.5 mm thick M.S. sheet, front glass locking arrangement supported on M.S. structural members, painting with two coats of postal red enamel.

2. HOSES AND HOSE CABINET

All hoses shall be numbered and a record submitted with completion plane. The number and length shall be easily recognizable on each hose pipe.

External hose boxes shall be installed such that the hose is not exposed to sun rays.

3. PAINTING

Painting of the entire wet riser piping over the ground shall be done with anticorrosive primer and 2 coats of approved paint. The color shall be red to shade No. 536 of IS: 5, Paint shall conform to IS:2932.

The pumps and engine shall be painted after installation with a coat of approved paint to similar shade as per original supply.

4. TESTING OF THE SYSTEM

After laying and jointing, the entire piping shall be tested to hydrostatic test pressure. The pipes shall be slowly charged with water so that the air is expelled from the pipes. The pipes shall be allowed to stand full of water for a period of not less than 24 hours and then tested under pressure. The test pressure shall be 12 kg/cm². The test pressure shall be applied by means of manually operated test pump or by a power driven test pump to be provided by the contractor. In either case precautions shall be taken to ensure that the required test pressure is not exceeded.

The open end of the piping shall be temporarily closed for testing.

Test shall be conducted on each pump set after completion of the installation with respect of delivery head, flow and B.H.P. The test shall be carried out by the contractor at his own cost.

All leaks and defects in different joints, noticed during the testing and before commissioning shall satisfy the satisfaction of the engineer.

Testing of fittings/ equipments shall be carried out either at site or at works in the presence of a representative of the engineer. Test certificates shall also be furnished by the contractor.

The automatic operation of the system for the various functional requirements and alarms as laid down in his specification shall be satisfactorily carried out on pressure of the engineer.

5. APPROVAL BY LOCAL BODIES

It shall be the responsibility of the contractor to obtain the approval of drawings and to get the installation inspected and approved by the concerned authorities as may be necessary as per local by laws, any fee payable to the local bodies for such activities shall also be borne by the owner on production of receipts for money paid and the other expenses will be borne by the contractor.

6. PIPE WORK ASSOCIATED WITH DIESEL ENGINE

Pipe works for fuel system, lube oil system and exhaust system shall be complete with all required supports, clamps, hangers etc. for a complete work.

Fuel feed is by gravity and the fuel tank shall be located at least 60cm above the fuel injection pump.

Fuel pipe of copper shall not be soldered but brazed or welded.

No valves or cocks shall be provided in the fuel feed line to engine from the fuel tank.

Precautions shall be taken to prevent any air locks in any part of the fuel system. No air relief cock shall be permitted and where inescapable, screwed plugs shall be provided for the purpose.

The installation of the fuel supply system shall be such that a completely primed condition is maintained, free from air lock.

Filters shall be provided in fuel oil and lube oil circuits allocations that are easily accessible for maintenance.

7. WET RISER PIPE WORK

The suction line for each pump shall be independent.

No sluice valve shall be provided in situation line, where the pump is located above the water level in the sump foot valve and strainer shall however be provided.

Butterfly or Sluice valve shall be provided in situation line, where the pump is located below the water level in the sump, strainer at the suction end shall be provided.

Each external (yard) hydrant shall be controlled by a Butterfly or sluice valve at ground level.

Butterfly or Sluice valves shall be kept in open position and the scope of work includes provision of necessary leather strap and pad lock so as to prevent unauthorized closing of valve.

The installation work includes provision of all clamps, supports, anchors etc.

Spacing between vertical supports shall not exceed 1.5m and horizontally at 2m up to 50mm and 1.5m for higher diameters. Clamps shall be provided on either side of the tee joints for internal hydrants. Necessary anchors/ thrust pads shall be provided as approved at locations of bends, tees etc. as required within the scope of work.

Underground pipes of the wet riser system shall be laid 1m below ground level and at least 2m away from the face of the buildings. The run of piping shall be preferably along roads and footpaths and shall not be under buildings. Where specifically indicated to cross buildings, these shall be laid in masonry trenches with removable covers. With cut off valves at the entry and exit points.

1.09 TECHNICAL SPECIFICATIONS FOR SPRINKLER SYSTEM

1 SPRINKLER HEADS

- a. Sprinkler heads shall be of quartzoid bulb type with bulb, valve assembly yoke and the deflector. The sprinklers shall be of approved make and type.
- b. Types
 - i. Conventional Pattern
The sprinklers shall be designed to produce a spherical type of discharge with a portion of water being thrown upwards to the ceiling. The sprinklers shall be suitable for erection in upright position or pendant position.
 - ii. Spray Pattern

The spray type sprinkler shall produce a hemispherical discharge below the plane of the deflector.

iii. Ceiling (flush) Pattern

These shall be designed for use with concealed pipe work. These shall be installed pendant with plate or base flush to the ceiling with below the ceiling.

iv. Side Wall Sprinklers

These shall be designed for installation along with the walls of room close to the ceiling. The discharge pattern shall be similar to one quarter of sphere with a small proportion discharging on the wall behind the sprinklers.

c. Constructions

- i. Bulb:- Bulb shall be made of corrosion free material strong enough to withstand any water pressure likely to occur in the system. The bulb shall shatter when the temperature of the surrounding air reaches a predetermined level.
- ii. Valve Assembly: - Water passage of the sprinkler shall be closed by a valve assembly of flexible construction. The valve assembly shall be held in position by the quartzoid bulb. The assembly be stable and shall withstand pressure surges or external vibration without displacement.
- iii. Yoke: - The yoke shall be made of high quality gun metal. The arms of yoke shall be so designed as to avoid interference with discharge of water from the deflector. The sprinkler body shall be coated with an approved anti-corrosive treatment if the same is to be used in corrosive conditions.
- iv. Deflector: - The deflector shall be suitable for either upright or pendent erection. The deflector shall be designed to give an even distribution of water over the area protected by each sprinkler.

d. Color Code

The following color code shall be adopted for classification of sprinkler according to nominal temperature ratings:

Sprinkler Temperature Rating	Color of the Bulb
57 deg. C	Orange
68 deg. C	Red
79 deg. C	Yellow
93 deg. C	Green
141 deg. C	Blue
182 deg. C	Violet/ Mauve
204/260 deg. C	Black

e. Size of Sprinklers Orifices

The following sizes of sprinklers shall be selected for various classes or hazards.

Extra light hazard	10/15 mm nominal bore
Ordinary light hazard	15 mm nominal bore
Extra high hazard systems	30 sprinklers

f. Stock of replacement sprinkler

The following spare sprinklers shall be supplied along with the system.

Extra high hazard systems	6 sprinklers
Ordinary hazard systems	24 sprinklers
Extra high hazard systems	36 sprinklers

g. Temperature Rating

For normal conditions in temperature climates rating of 68/74 deg. C shall be used. However, the temperature rating shall be as closed as possible to, but not less than 30 deg. C above the highest anticipated temperature conditions.

2 PIPES AND FITTINGS

- a. Pipes for wet riser system shall be black steel conforming to IS: 1239 (Heavy Class)

Fittings for black steel pipes shall be malleable iron suitable for welding or approved type cast iron fittings with tapered screwed threads.

Jointing

Joint for black steel pipes and fittings shall be metal to metal tapered thread or welded joints. A small amount of red lead may be used for lubrication and rust prevention in threaded joints.

Joints between MS pipes, valves and other appurtenances, pumps etc. shall be made with M.S. flanges with appropriate number of bolts. Flanged joints shall be made with 3mm thick insertion rubber gasket.

Pipe Protection

All pipes above ground and in exposed locations shall be painted with one coat of red oxide primer and two or more coats of synthetic enamel paint of approved shade.

Pipes in chase or buried underground shall be painted with two coats of hot bitumen, wrapped with bituminous Hessian cloth and finished with one coat of hot bitumen paint.

Pipe Supports

All pipes shall be adequately supported from ceiling or walls from existing inserts by structural clamps fabricated from M.S. structurals e.g. rods, channels, angles

and flats. All clamps shall be painted with one coat of red and two coats of black enamel paint. Where inserts are not provided, the contractor shall provide anchor fasteners.

Orifice Flanges

Contractor shall provide orifice flanges fabricated from 6mm thick stainless steel plates on the branch lines feeding different zones/ floors so as allow required flow of water at 3.5 kg/ sq.mm pressure. The contractor shall furnish design for these orifice flanges.

Valves

Butterfly or Sluice valves of size 80mm and above shall be double-flanged cast iron conforming to IS: 780.

Check valve shall be of cast iron double flanged conforming to IS: 5312.

Valves on pipes 65mm and below shall be heavy pattern gunmetal valves with cast iron wheel seat tested to 20 kg/ sq.mm pressure. Valves shall conform to IS: 778.

Air Valves

25mm dia screwed inlet cast iron single acting air valves on all high points in the system or as shown on drawings.

Drain Valves

50 mm dia black steel pipe conforming to IS: 1239 heavy class with 50 mm gunmetal full way valve for draining water in the system in low pockets.

- 3** **INSTALLATION CONTROL VALVE:-** Installation control valves shall comprise of the following.
 - a. One main stop valve of full way pattern with gunmetal pointer to indicate where open/ shut
 - b. One automatic alarm valve, fitted with handle and cover.
 - c. One hydraulic alarm motor and gong for sounding a continuous alarm upon out-break of fire.
 - d. One combined waste and testing valve including 5 mtr of tubing and fittings
 - e. Alarm stop valve
 - f. Strainer
 - g. Drain plug
 - h. Padlock & strap
 - i. Wall box for installation of valve

- 4** **PRESSURE GAUGES: -** Burden type pressure gauges conforming to IS/ BS specifications shall provided at the following locations.
 - a. Just above alarm valve
 - b. Just below alarm valve, on the installation stop valve
 - c. One pressure gauge on delivery side of each pump

- d. Required number of pressure gauges on pressure tank

5 INSTALLATIONS OF PIPING

- A. Below ground piping: -Underground piping should be installed in masonry trenches with cover or reinforced concrete. The pipe work shall be supported at regular intervals of 2.5m with masonry or RCC supports. Wherever pipes pass through roads/ pavements shall be protected against corrosion with two coats of bituminous painting and wrapped with pypkote or bitumen Hessian cloth and finish with one coat of hot bitumen paint.

A. ABOVE GROUND PIPING: -

- a. All above ground piping shall be installed on suitable to pipe hangers/ supports as required. The hangers shall be made of MS angles, channels, channels etc. and painted to the required finish (with suitable synthetic enamel paint). The spacing supports shall be as follows.
- | | | |
|------|---------------------|----------|
| i. | 20 mm to 32 mm dia | 2 mtr |
| ii. | 40 mm to 65 mm dia | 2 mtr |
| iii. | 65 mm to 100 mm dia | 1.75 mtr |
| iv. | above 150 mm dia | 1.50 mtr |
- b. Piping shall be so installed that the system can be thoroughly drained. All the pipes shall be arranged to drain to the installation drain valve. In case of basement and other areas where the pipe work, is below the installation drain valve / auxiliary valves of the following sizes shall be provided.
- | | |
|------|--|
| i. | 20 mm dia valve for pipes up to 50mm dia |
| ii. | 25 mm dia valve for 65 mm dia pipes |
| iii. | 32 mm dia valves for pipes larger than 65 mm dia |
- c. Piping shall be screwed type up to 50 mm dia. Welding of joints will be allowed for pipes of 50 mm of larger diameters.
- d. The piping shall be pressure tested by the hydrostatic method upto a pressure of 1.5 times the working pressure the piping shall be slowly charged with water so that all the air is expelled from the piping by providing a 25mm inlet with a stop cock. The piping shall be allowed to stand full of water for a period of 2 hours and then the piping shall be put under pressure by means of manually operated test pump or by a power driven test pump. The pressure gauges used for testing shall be accurate and shall preferably be calibrated before the testing is carried out. All the leakages and defects in joints revealed during the testing shall be rectified to the entire satisfaction of the consultant. The system may be tested in sections parts as the work of erection of piping proceeds. The piping shall withstand 1.5 times the working pressure for at least 2 hours.

7 ANNUNCIATION SPRINKLER PANEL

The equipment for control panel should be compact neatly wired and enclosed in a suitable 2 mm M.S. sheet that is suitably treated against corrosion. The control panel should be painted with enamel paint. The panel shall consist of:

- a. Panel should be made in a module of 20 zones e.g. Each module will have audible and visual indications and will monitor the circuit conditions. With 24v DC battery.
 A.C. Power Supply
 Fault and Fire indication lamp
 Alarm acknowledgment push buttons
- b. The circuits provided in the control panel for each zone shall indicate the following conditions:
 - i. Open Circuit in zone wiring
 - ii. Short Circuit in zone wiring
 - iii. Normal conditions
 - iv. Power failure
 - v. Low battery
- c. The Automatic annunciation panel shall suitable for operation on 24V DC and shall be provided with power supply unit suitable to operate on A.C. mains of 230 V with a variation of 10%. The system shall be so designed that in case of failure of A.C. main supply it shall automatically change over to battery supply.
- d. Suitable protection may be provided against charging of the battery over and above the specified values.

8 BATTERY UNIT

- i. The system shall be powered by lead acid storage stationery complete with automatic dual rate charger boost and trick operating form 220 V, 50 Hz, single phase, mains supply. The battery capacity should be adequate for operation of the system connected to it for at least 24 hours in the non-alarm state followed by 30 minutes operation of all sounders and other connected equipments after a power (mains) failure.
- ii. The automatic charger should operate at the boost charge when the battery terminal voltage is less than about 2.1 V 20 per cell, and operate at a trickle charge rate of 100 to 200 AH, when the battery terminal voltage exceeded about 2.25 per cell.
- iii. The power unit should have the following.
 - a. Voltmeter 0-03 V
 - b. Ammeter of suitable range
 - c. Indicator lights for mains
 - d. Indicator lights for DC output
- iv. The preferred nominal DC voltage shall be 24 V and shall preferably be isolated. (IF and isolated supply is provided a line earthing indicator should also be provided).
- v. The DC system and the detection and sounder circuits shall be protected against their attaining a voltage to earth exceeding 50V.
- vi. The connection to the 220 V, 50 Hz, single phase system shall be through a three pin plug socket especially provided for the connection to the annunciation panel. This connection should in addition utilized for earthing all non-current carrying metal parts of the sprinkler system, except those that are either doubly insulate or mounted at a height exceeding 2.2 meters.

- vii. The battery unit shall be housed in a steel cabinet at least 2 mm thick suitably painted with two coats of Post Office Red, Enamel necessary vent holes should be provided for proper ventilation.

1.10 STANDARDS AND CODES

1. IS 1648 Code of practice for fire safety of building (general) firefighting equipment and maintenance.
2. IS 3844 Code of practice for installation of internal fire hydrant in multistory buildings
3. IS 2217 Recommendations for providing first aid and fire fighting arrangement in public buildings.
4. IS 2190 Code of practice for selection, installation and maintenance of portable first aid fire appliances.
5. Part IV, fire fighting National building code
6. IS 5290 External fire hydrants
7. IS 5290 Internal landing valves
8. IS 904 2 & 3 way suction collecting heads
9. IS 884 First aid hose reel
10. IS 5132 High pressure rubber pipe
11. IS 1537 C.I. Double flanged pipes
12. IS 1538 C.I. Double flanged fittings
13. IS 780 C.I. Sluice valves and gunmetal valves
14. IS 6234 Specifications for Water type (stored pressure) fire extinguisher.
15. IS 2873/2171 Specifications for fire extinguisher of Carbon-di-oxide & Dry powder type.

ANNEXURE-II

TECHNICAL SPECIFICATIONS FOR INTELLIGENT FIRE DETECTION & ALARM SYSTEM

SECTION-1

General & System Engineering:

- 1.1 An Intelligent Addressable Fire Alarm System shall have/ able to provide the following basic features as under:
 - (i) Accurate quick response to emergencies.
 - a. Early fire warning
 - b. Individual Device identification / location
 - c. Analog Signal feed back
 - d. Pre-alarm notification
 - (ii) Eliminates nuisance alarms
 - (iii) Powerful operator interface
 - (iv) Reliable
 - (v) Advance on line diagnostics i.e.
 - a. Continuous circuit supervision
 - b. Detection and isolation of faults
 - c. Continuous processing of analog value sensor data
 - d. Maintenance diagnostics
 - e. Panel test
 - f. Control Board self test
 - (vi) Reduction in maintenance costs by:
 - a. Remote testing and sensitivity adjustment.
 - b. Panel testing of smoke sensor to meet International Standard.
 - c. Individual sensor cleaning on need based only.
 - (vii) Flexible application
- 1.2 An intelligent addressable fire alarm system is provided to effect total control over the life safety services required in the building. The system is provided with addressable and analog fire alarm initiating, annunciating and control devices. The system is such that smoke sensors/detectors, thermal sensors/detectors and manual call points etc. can be identified with point address.
- 1.3 An intelligent addressable fire alarm system shall be capable of:
 - (a) Setting smoke sensor sensitivity remotely on a pre-programmed sequence. The system is able to recognize normal and alarm values that reveal trouble condition, pre alarm or need of maintenance.
 - (b) Provide a maintenance / pre-alert capability at smoke sensors to prevent the detectors from indications a false alarm due to dust, dirt, etc.
 - (c) Provide alarm verification of individual smoke sensors. Alarm verification printed on the in-built printer of the main fire alarm panel on the external printer attached.

- (d) Provide outputs that are addressable i.e. output have a point address to be set from the main control station. The operator is able to assign such points to logical point groups (software zones) for pre-programmed operation.
- (e) In the event of fire alarm is notified but not in a fault condition the following activity/ command shall be initiated by the FACP:
 - i. The exact location is shown on the main fire alarm panel,
 - ii. Local sounder automatically sounded
 - iii. The complete information is printed along with time and date of occurrence (event log).
 - iv. Air handling units on the effected zones is automatically switched OFF and simultaneously respective fire dampers are closed.
 - v. Pressurization fans on the evacuation shafts i.e. stairwells, etc. are automatically switched ON.
 - vi. All system output program assigned via control-by event programs that are activated by a particular point in the alarm are executed
 - vii. The associated system output (alarm notification appliance and/ or relays) are activated.
 - viii. The audio portion integrated to the system directs the proper signal (tone or Voice) to the appropriate speaker circuits. Pre-recorded alarm messages are played on the public address system.
- (f) All the loop elements are able to isolate the short circuits on the loop wiring and thus no element is isolated in the event of single short circuit and the exact location of the short circuit is indicated on the respective fire alarm panel.
- (g) The fire alarm panels have necessary interface units for intelligent addressable detectors alarm output modules for external actuations through fail safe programmable relays and communication interface.
- (h) Fire alarm management system in an intelligent addressable Fire Alarm and Detection System shall comprise Graphical Management System which is multi-user, with a lot of facilities. It shall have two screen display, one screen dedicated to display the graphic (building floor plans, Alarm panels etc.) and the another screen lists all events (alarm reports , Fault messages, information about disable detectors etc.)
- (i) Building management system, if any, provides bidirectional monitoring and operation of the attach devices of the Fire Alarm and Detection Systems. Besides, other components/ devices in the building like AHU's Pressurization Doors , lifts etc. can also be monitored and operated from the graphical fire alarm management software.

SECTION-2

MANUAL CALL POINTS

2.1 Scope

This section covers the requirements of manual call points (MCP) used in addressable intelligent fire alarm systems.

2.2 Constructional requirements of:-

- i. It is used to trigger a fire alarm manually, and should be suitable for connection to the system loop technology. The alarm is raised by breaking the glass with the alarm carrying on until a replacement glass panel is fitted or alarm to be made silent from control panel. Suitable arrangement like scratching by a diamond bit shall be incorporated in the frangible element.
- ii. A micro switch within the units is held “off” by the edge of the glass when glass is broken the switch is released and a signal is transmitted to control panel where the alarm is raised. The glass is preferably plastic coated to eliminate the damage of splinters causing injury.
- iii. For testing without breaking the glass a test key, should be available so as to lower or remove the glass to release the micro switch. Full functional test is achieved accordingly.
- iv. Facility for reset/adjustment of manual call point alarm triggering.
- v. The unit shall be made out of polycarbonate ABS blend with flame retardant & self extinguishing properties or cast aluminum or ASA plastic or as per manufacturer design and applicable standards.
- vi. It shall preferably contain an integrated short circuit isolator which should ensure that a fault is localized and that the loop continues to function fully in the event of a wire break or a short circuit.
- vii. The call box shall have suitable provision to knock out of termination of a 20mm conduit. This shall also have suitable provision for being fixed on surface or semi recessed in wall.
- viii. The word 'FIRE' shall appear embossed in red color on the front.
- ix. Acknowledgement LED displays – red for triggered alarm indication (flashing) and / or maintenance evacuation.
- x. Protective category shall be IP54 for indoors & IP65 for outdoors applications.

SECTION-3

INTELLIGENT ADDRESSABLE FIRE DETECTORS

3.1 Scope

The section covers the requirement of automatic intelligent addressable fire detectors.

3.2 Type of Detectors

The detectors shall be one or more of following types:

(A) Heat Detectors:

- (i) Fixed Temperature
- (ii) Rate of Rise-Cum-fixed temperature Detectors
- (iii) Probe type high temperature Bi-Metal Heat Detector
- (iv) Linear Heat Sensing Cables

(B) Smoke Detectors:

- (i) Ionization type
- (ii) Optical type
- (iii) Air Sampling type (Laser type Detector or high sensitivity detector)
- (iv) Multi-criteria Type
- (v) Photo-thermal Type
- (vi) Beam Detector
- (i) Carbon Monoxide

(C) Spark / Ember Detector

(D) Ultra Violet (UV) Flame Detector

(E) Infra Red (IR) Flame Detector

(F) Thermal Multi-Criteria Detector

3.2.1 Governing Specifications:

- (i) Heat Detectors shall conform to IS-2175 - 1988 or BS-5445 (EN 54) Part 5 – 1977(VdS/LPCB approved) or NFPA72 (UL/ULC/FM approved) amended upto date.
- (ii) Smoke Detectors shall conform to IS-11360-1985 or BS 5446 Part I-1977 & Part VII –1985 (VdS/LPCB approved) or NFPA72 (UL/ULC/FM approved) amended upto date.
 - (ii) For other detectors/devices relevant IS or BS (EN 54) (VdS/LPCB approved) or NFPA 72 (UL/ULC/FM approved) amended upto date.

3.2.2 Detectors & Applications:

i. Fixed Temperature Heat Detector:

The fixed temperature heat detectors are designed to operate when the temperature of detectors exceeds a predetermined value.

Application : Used where the ambient temperature is likely to fluctuate rapidly over short period e.g. in kitchen, boiler room, D.G. Set room, Non AC. area in building.

ii. Rate of rise-cum-fixed temperature detectors:

These detectors are designed to operate within a given time.

- When the rate of temperature rise at the detector exceeds a predetermined value regardless of the actual temperature.
- When temperature at detector exceeds a predetermined value.

Application -

In areas subject to smoke, dust and dirt during normal use, where smoke detectors cannot be used.

iii. Probe type High Temperature Bi-Metal Heat Detector:

The detector are reset type and highly suitable to use above 80°C where electronic components cannot be used.

Application: Generator Enclosure, turbine enclosures, oven and furnace area etc.

iv. Linear Heat Sensing Cables:

These can be broadly divided into two categories; digital and analogue, depending upon the principle by which the sensing cable registers a change in temperature.

Digital sensor consists of two core cable in which the conductors are separated by a heat sensitive insulator. When a specified temperature is reached, the cable insulation breaks down and an alarm is activated.

In analogue sensor, cores are separated by a negative temperature co-efficient polymer whose resistance reduces in proportion to temperature increase.

Applications: These cables are used for detecting fire and overheating in following areas:

- (i) Cable tunnels, trays and vaults
- (ii) Material conveyors
- (iii) Bulk storage mull - racked areas
- (iv) Rim seals of floating roof tanks storing hazardous chemicals.

v. Ionization smoke Detector

Good for invisible smoke detection. Due to use of radioactive element present in it, the disposal is a great problem and can cause serious health hazard, hence now slowly becoming obsolete, use of this detector may be avoided.

vi. Optical Smoke Detectors

The optical smoke detector is based on Light Scattering principle. The LED (Light Emitting Diode) transmits light to the measuring chamber where it is absorbed. In case of fire, smoke enters the measuring chamber and the smoke particle scatter the light. The amount of light reaching to photo diode is converted into a proportional electrical signal. On operation of detector when the electrical signal reaches a predetermined threshold value it triggers the alarm on control panel.

Use: More suitable for smoldering fire which produces mainly larger particle of smoke e.g. burning of PVC, in A.C. area in a building.

vii. Air sampling type detector (High sensitivity smoke detector)

Laser type smoke detectors are used in this type of system. Detector consists of a piping or tubing distribution network that runs from the detector to the areas to be protected. An aspiration fan in the detector housing draws air from the protected area back to the detector through air sampling ports, piping or tubing. At the detector, the air is analyzed for the products by the laser based smoke detector.

Use : Typical application of the system is in places where trace of smoke needs to be detected and high airflow can make traditional smoke detector inadequate.

viii. Spark / Ember Detector

This detector uses a solid state photodiode or photo transistor to sense the radiant energy emitted by embers typically between 0.5 microns and 2.0 microns in normally dark environments.

These detectors can be made extremely sensitive (microwatts) and their response times can be made very short (microseconds).

Applications :

The detectors are primarily installed to detect sparks and embers that could, if allowed to continue to burn, can cause a much larger fire or explosion.

The detector is mounted on some form of duct or conveyor, monitoring the fuel as it passes by. It is necessary to enclose the portion of the conveyor where the detectors are located, as these devices generally require a dark environment.

ix. Ultra-Violet (UV) Flame Detector:

It makes use of ultra violet sensitive photocathode for detecting flame. It has very high sensitivity and low false alarm rate and is suitable for indoor application.

The detector is not suitable for high ceiling as the radiation from fire is attenuated by smoke. The random UV radiation from sources such as lighting, the Arc welding etc. can cause false alarms hence usage in such areas to be avoided.

x. Infra-red flame detector:

Almost all the materials that participate in the flaming combustion emit ultraviolet radiation to some degree during flaming combustion whereas only carbon containing fuels emit significant radiation at 4.35 micron (Carbon dioxide) band used by many detectors to detect a flame. This detector senses wave length in the infrared spectrum.

Application:

- (i) High-ceiling, open-spaced buildings such as ware-houses and aircraft hangers
- (ii) Outdoor or semi outdoor areas where winds or draughts can prevent smoke from reaching a heat or smoke detector.
- (iii) Where rapidly developing flaming fire can occur such as in petrochemical production areas, storage areas, natural gas installations, paint shops or solvent areas.

Some of the extraneous sources of radiant emissions that have been identified as interfering with the stability of flame detectors include the following:

- (a) Sun Light
- (b) Lightning
- (c) X-rays
- (d) Gamma Rays
- (e) Cosmic Rays
- (f) Ultraviolet radiation from welding
- (g) Electromagnetic interference
- (h) Hot objects
- (i) Artificial lighting

xi. Photo - Thermal Multi-criteria detector

This detector combines optical smoke and heat detection and is capable of being used as a smoke and heat detector both. The multi criteria detector shall be able to discriminate between non-threatening deceptive phenomenon such as cigarette smoke and actual fire hazards. This avoids nuisance alarm.

Application: This is good for smoldering and fast flaming fires, optically dense smoke and can be used in buildings for AC and Non-AC area.

3.3 General Features common to all Detectors:

- 3.3.1 Built-in-response Indicator: Each detector shall incorporate indicator "LED" at the detector which shall blink on actuation of the detector to locate the detector which is operated while on fire. The detector shall not be affected by the failure of the response indicator lamp.
- 3.3.2 The detector shall have preferably 360 visible view of the alarm LED or twin LEDs blink whenever detector is addressed.
- 3.3.3 The detector shall have preferably with integrated built in short circuit isolator or as per manufacturer design so that in case of any short circuit in the detector, the detector can be isolated so that loop should be able to retain the full functionality or in case built in short circuit isolator not available/ provided by the manufacturer a fault isolator unit shall be provided after every 20 detectors/ devices in a loop. But in later case isolator unit shall ensure that the loop is functional except the faulty detector.

- 3.3.4 The detector shall be with IP 54 protection category with base.
- 3.3.5 The detector shall be with built in microprocessor capable of making alarm on control panel based on the information stored in detector.
- 3.3.6 The detector shall confirm to international standard i.e. NFPA/EN or equivalent.
- 3.3.7 The detector shall provide electronic address setting by means of configuration software.
- 3.3.8 The detector shall be plug-in-type and shall have common base to facilitate exchange for cleaning & maintenance. The base of the detector shall be interchangeable with other smoke detectors.
- 3.3.9 The detector shall be supplied fully tested and each detector should bear the SI. No. and seal of the approving laboratory/body.
- 3.3.10 The detectors, device including response indicator and control panel shall be of same make.
- 3.3.11 Reversed polarity or faulty zone wiring shall not damage the detector.
- 3.3.12 The detector shall be suitable for column/ceiling mounting.
- 3.3.13 The detector should have static voltage over load protection.
- 3.3.14 The detector shall be individually identifiable from the control unit by geographical location in the system.
- 3.3.15 The detector shall connect to the local control unit via a fully supervised two wire circuit.
- 3.3.16 The detector shall be inserted into or remove from the base by a simple push twist mechanism with an appropriate tool.

3.4 Addressable Beam Detector (Infra-red-light detector)

- (i) The Beam Detectors shall be long range, projected beam type smoke detectors which consist of a separate transmitter and receiver and evaluating unit integrated into a compact housing.
- (ii) The transmitter shall emit an invisible pulsed infra red beam to receiver when beam is obscured beyond selected threshold (20%, 30%,.....70% obscuration) by smoke, the receiver shall signal an alarm. If the beam is completely blocked, receiver shall signal a trouble.
- (iii) It shall provide selectable sensitivity and alarm response time for flexibility of installations.
- (iv) It shall have a range coverage from 9 mtrs. to 100 mtrs. with upto 16-17 mtrs. Spacing between adjacent detectors on smooth , flat ceilings.
- (v) It shall have automatic environment compensation.

3.5 Addressable Heat Detector:

Type of Heat Detectors shall be as per para no. 3.2.2 – i. - iv. in addition the following features shall also be provided:

- (i) The detector and transmission electronics in the detector head for easy exchange in the event of malfunction. Therefore no electronics are permitted in base.
- (ii) It has maximum triggering temperature programmable from 58 degree C to 81 degree C.
- (iii) It has high level of immunity to electromagnetic influences.
- (iv) The alarm signal when limit temperature is reached as per (ii) above.
- (v) Automatic addressing during initial set up or exchange of the detector, without making adjustment of detector.
- (vi) Adjustment of the maximum temperature value in combination with or without the temperature differential value.
- (vii) Theft protection to protect against unauthorized removal of the detector from base can be available.

3.6 Addressable Photo (optical) - Thermal (Multi Criteria Detector):

- 1. The intelligent multi criteria detector shall have photo electric and thermal technologies in a signal sensing device.
- 2. The device shall include to combine the signal of the thermal sensor with the signal of the photoelectric signal in an effort to react in the event of a fire condition.
- 3. It shall have an ability to distinguish between a fire condition and a fire alarm condition by examining the characteristics of the thermal and smoke sensing chambers and comparing them to database of actual fire and deceptive phenomena.

Use:

- (i) These detectors shall be utilized for slow smoldering fire.
- (ii) The microprocessor shall be capable of selecting the appropriate sensitivity levels based on the environmental conditions, it can be in office, manufacturing unit, kitchen etc. and have the ability to automatically change the setting as the environment changes.

3.7 Intelligent Duct Smoke Detectors

- (i) The smoke detectors housing shall accommodate either an intelligent ionization detector or an intelligent photo electric detector of that provides continuous analog monitoring and alarm verification from the panel through a control module in the loop system.
- (ii) When sufficient smoke is sensed in the return air coming to AHU (Detector unit has to be installed in return air path) an alarm signal is initiated at FACP to close the fire damper and prevent the rapid distribution of toxic smoke and fire gases throughout the areas served by duct system.

Note : Various types of detectors their use and applications are same as per S.n.-3.2.2 above.

3.8 Detector Base:

- (i) The detectors of all types shall fit into a common type of standard base.
- (ii) Once a base has been installed, it shall be possible to insert, remove and exchange different types of detectors by a simple push & twist movement.
- (iii) The standard base shall be equipped with screw less wiring terminals suitable for securing wire size upto 2.5 sq.mm. and with built in strain limits to prevent permanent terminals deformation and weakening of contact pressure.
- (iv) It shall have a sealing plate to prevent dirt, dust condensation or water from the conduit reaching the wire terminals of the detector contact points.
- (v) All standard base shall be supplied with a removable dust cover to protect the contact area during installation and construction phase of the building. It must allow the inspection and verification of the zone wiring before insertion of any detectors.
- (vi) The standard base shall have a built in mechanism, which allows mechanical locking of any installed detector head, thus preventing unauthorized removal or tempering while maintaining.
- (vii) Reversed polarity or faulty zone wiring shall not damage the detector.

3.9 Intelligent Addressable Monitor Module

- (i) The module shall be suitable to monitor the polling of up to minimum four potential free contacts by using a single or multiple housing and designed for surface or flush mounting.
- (ii) The operating mode can be selected and is assigned separately for each input the addressing of the module and the setting of parameters should be carried out using PC software, via the fire alarm control panel.
- (iii) The module should contain an integrated short circuit isolator which ensures that the fault is localized and the loop continues to function fully in the event of a wire break or a short circuit.
- (iv) The module should be minimum IP 40 protection category when housed in a case or as per manufacturer design.
- (v) The module should be with soft addressing to be set with a PC via fire alarm panel at the time of programming.
- (vi) The module should have integrated buzzer force signaling maintenance alarms.
- (vii) An LED shall be provided that shall flash under normal condition indicate that module is operational.

3.10 Intelligent Addressable Relay / Control Module

- (i) Relay module may contain upto four relays each with a potential free changeover contact.
- (ii) Each relay contact should be FAIL/ SAFE programmable, means it should be possible to program each contact to operate in three conditions viz. 'NO FIRE CONDITION' , 'FIRE CONDITION' and 'LOOP POWER FAILURE CONDITION'.
- (iii) Both the addressing of the module and setting of parameters to be carried out using PC software via the fire alarm control panel.
- (iv) The module contains a short circuit isolator, which ensures that the fault is localized and that the loop continues to function fully in the event of a wire break or a short circuit.
- (v) The module should be minimum with IP 40 protection category when housed in a case or as per manufacture design.
- (vi) It should have integrated buzzer for signaling maintenance alarms.
- (vii) Its housing have is designed for surface or flush mounting.

3.11 Isolator Module:

- (i) The module shall limit the number of module of detectors that may be rendered inoperative by a short circuit on the loop segment.
- (ii) At least one isolator module shall be provided for each floor or protected zone of the building.
- (iii) If a wire short circuit occurs, the isolator module shall automatically open circuit (disconnect) the loop. When the short circuit condition is corrected, the module shall automatically reconnect the isolated section.
- (iv) The module shall not require any address setting and its operation shall be automatic. It shall not be necessary to replace or reset an isolator module after its normal operation.
- (v) It shall give information on the control panel in the form of audio- visual alarm enunciations along with list of detectors which have become non-functional.

SECTION-4

INSPECTION, TESTING AND COMMISSIONING

8.1 SCOPE:

This section covers the requirements for testing and commissioning of all types of fire alarm systems/ Automatic Fire Alarm System/ Intelligent addressable fire alarm system.

8.2 Testing before supply

- (i) Detectors, control and indicating panels, sounders, PA system equipment and battery unit with charger etc. shall be tested at the manufacturer's works to indicate satisfaction to the contract specifications, and test certificates be furnished with the supply at site of work by the contractor.
- (ii) In case of imported fire alarm components such as main fire alarm panel, Repeater panel, Smoke, Heat and Beam detectors, Manual Call points, Monitor and Control modules, Response Indicators, etc; individual certificate with individual number confirming to the relevant international standard shall be provided by the contractor at the time of supply at site of work.

8.3 Initial Installation Inspection:

- 8.3.1 After installation, a visual inspection of all the detectors should be made to make sure that they are properly placed. Each detector should be inspected to ensure that it is properly mounted and connected.
- 8.3.2 Restorable heat detectors and restorable elements of combination detectors should be tested by a heat source, such as a hair dryer, or shielded heat lamp until it responds, making sure that the sensing element is not damaged. After each heat test, the detector should be reset precaution should be taken to avoid damage of the non- restorable fixed temperature element of a combination rate of rise /fixed temperature detector.
- 8.3.3 Non-resettable fixed temperature heat detectors which are not to be heat-tested should be tested mechanically or electrically for fire alarm function.
- 8.3.4 Heat detectors with replaceable fusing alloy element should be tested first by removing the element to see whether contact operate properly and then reinserting them in proper position.
- 8.3.5 In periodic testing, heat detectors should be visually examined for damaged for other conditions (such as heavy coats of paints etc. likely to interfere with the correct operation)
- 8.3.6 Each smoke detector should be tested to initiate an alarm at its installed location with smoke or other approved aerosol which demonstrates that the smoke can enter the chamber and initiate an alarm.

- 8.3.7 In order to ensure that each smoke detector is within its sensitivity range, its sensitivity range, it should be tested using either:
- a) A calibrated test method, or
 - b) A manufacturer's supplier's approved calibrated sensitivity test instrument, or
 - c) Approved control equipment for the purpose, or
 - d) Other approved calibrated sensitivity test method.

8.3.8 Detectors found to have a sensitivity outside the approved range should be replaced.

Note : Detectors sensitivity cannot be tested or measured using any spray/ smoke producing device that administers an unmeasured concentration of aerosol/ smoke into the detector.

8.4 Testing after installation:

Following tests shall be conducted in a completed FAS/AFAS/intelligent addressable FAS installation, in the presence of the Engineer-in-Charge and the test certificate shall be furnished for the following test:

- (i) Continuity test.
- (ii) Test for insulation resistance of the wiring and the control and indication panels.
- (iii) Test for system operation- In the case of AFAS (see note below), this test for fire

conditions shall be conducted using a test fire at normal floor level. The number of locations for such a test shall be 1% of the total number of detectors in an installation subject to a minimum of two and maximum of five. The system operation for fault conditions shall be conducted by introducing faults such as open circuit, short circuit, removal of detector, open/short circuit in a sounder circuit etc.

Note: Use of a test fire in the case of installation using heat detectors need not be insisted upon.

8.5 Inspection by local bodies:

It shall be the responsibility of the contractor to get the installation inspected and passed by the local authorities concerned, as may be required by the local bye-laws, including payment of necessary inspection fee and an 'No Objection Certificate' (N.O.C) shall be obtained from the authority.

LIST OF REFERRED INDIAN STANDARDS

IS No.	Title
BS 7629-1 (Latest edition)	Fire Survival Cable - Copper conductor with glass mica fire barrier tape covered by an extruded layer of Cross-linkable Low smoke halogen free (LSZH) insulation
694: 2010	Polyvinyl chloride insulated unsheathed and sheathed cables/ cords with rigid and flexible conductor for rated voltages up to and including 450/750 V (fourth revision)
1255 : 1983	Code of practice for installation and maintenance of power cables upto and including 33 KV rating (second revision)
1554(Part-I): 1988	PVC insulated (heavy duty) electric cables: Part 1 for working voltage upto and including 1100 V (third revision)
2175 : 1988	Specification for heat sensitive fire detectors for use in automatic fire alarm system (Second revision)
9968(part 1) :1988	Elastomer insulated cables : Part 1 for working voltage upto and including 1100 V (first revision)
11360 : 1985	Specification for smoke detectors for use in automatic electrical fire alarm system.
2189 : 2008	Code of Practice for selection , installation and maintenance of Automatic fire detection and alarm system (second revision)- Code of Practice
NBC-2016	National Building Code, 2016

ANNEXURE-III

Technical Specification

EMERGENCY EXIT SIGNAGES

TECHNICAL SPECIFICATION

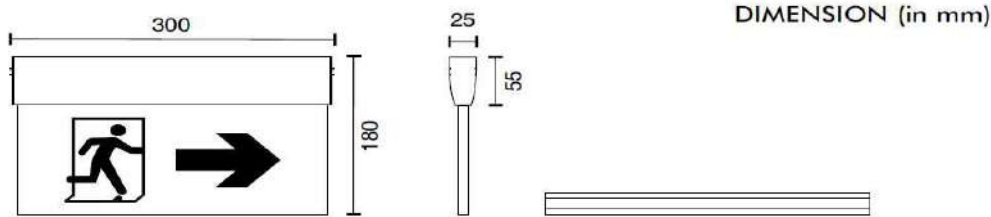
PORTABLE EMERGENCY LIGHT

Emergency Exit Signages shall provide illumination on normal mains power supply and the same lamp/s continues to provide illumination during mains failure on self-contained battery source. **ON** resumption of mains supply, the lamp/s switches back to the mains automatically & simultaneously starts recharging the battery to the required level.

Maintained models provide uninterrupted illumination.

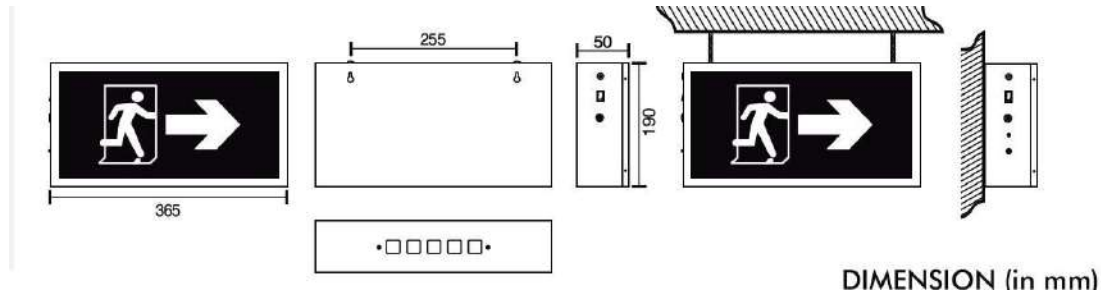
The dimensions of the signage shall be as given below with $\pm 15\%$ margin

a) For False Ceiling/ Low Ceiling Mounting:



CONFORMS TO	:	IS - 10322
I.P. (Ingress protection)	:	IP – 20
Housing alloy.	:	Made up of uniformly extruded, aluminum
Paint & Colour colour.	:	Epoxy polyester powder coating / White in
Diffuser cum message panel sheet.	:	CAD Designed printing on 8 M. M. clear acrylic
Input (V)	:	230 Volts, 50 Hz; $\pm 10\%$.
LED	:	Multiple LEDs.
LED Make	:	EDISON, Osram,.
Mains Connector grade.	:	5 A, 250 Volts, PCB Mount with electrical
Battery type	:	Ni-Cd / Ni-Mh; rechargeable.
Duration	:	180 MINUTES back up; $\pm 10\%$.
Charging Type	:	Constant current charger.
Recharging cycle	:	16 to 24 hours recharging period.
Mounting type	:	Ceiling / wall thru 2 Nos. of key holes.

For Wall or high ceiling mounting:



CONFORMS TO	:	IS - 10322
I.P. (Ingress protection)	:	IP – 20
Housing	:	22 MS C.R.C.A. Fabricated.
Paint & Colour	:	Epoxy polyester powder coating, White in colour.
Diffuser cum message panel	:	2 m.m. thick milky acrylic with vinyl pasted on it.
Input (V)	:	230 Volts, 50 Hz; $\pm 10\%$.
LED	:	Multiple LEDs.
LED Make	:	EDISON, Osram,.
Battery type	:	Ni-Cd / Ni-Mh; rechargeable.
Duration	:	180 MINUTES back up; $\pm 10\%$.
Charging Type	:	Constant current charger.
Recharging cycle	:	16 to 24 hours recharging period.
Mounting type	:	On Wall/ Ceiling thru 2 Nos. of key holes and Chain & Hook suspender.

LIST OF APPROVED MAKES OF MATERIAL FOR EMERGENCY EXIT SIGNAGES

S.NO.	MATERIAL	BRAND NAME
1.	EMERGENCY EXIT SIGNAGES	PROLITE / SKP AUTOMATION / SAFENESS QUOTIENT LIMITED / ILLUMINATION INDIA
2.	FIRE SURVIVAL CABLE	POLYCAB/ KEI/ FUSION / FRTEK

ELIGIBILITY CRITERIA

SL. NO	CRITERIA	DOCUMENTS REQUIRED FOR CONFIRMATION
1	<i>Bidder (sole)/Lead Consortium Partner/Original Equipment Manufacturer (OEM)</i> a) Should have been established and operating for a period of at least 05 years prior to the date of Tender.	For (a): - Certificate of Incorporation issued by the Registrars of Companies “OR” - Certificate of Registration by Registrar of Firms “OR” GST Registration in case of proprietary firms “OR” -Relevant Registration/Incorporation Certificate issued by Concerned State authorities in case of Foreign Firms with proof of address. Note: Bidders other than OEM, must enclose proper authorization certificate.
2	<u>Annual Turnover</u> <i>Bidder (sole)/Lead Consortium Partner/Original Equipment Manufacturer</i> Should have a minimum average annual turnover of 59,93,028/- for previous three financial years ending 31-03-2023	Audited annual accounts
3	<i>Bidder (sole)/All Consortium Partners</i> Should have PAN/ TAN/TIN/GST registration	a) Copy of registration certificates for Indian parties.
4	Technical Experience: <i>Bidder (sole)/Lead Consortium Partner/Original Equipment Manufacturer</i> Should have supplied and commissioned at least 02 similar type of works and same or higher capacity in the last 05 years (ending on 31-10-2023) for minimum of 8.32 lakhs of project value.	Copy of the Contract/Order & other documentary evidence for successful supply & commissioning to be submitted. List of order executed for similar item indication order no. & date, name and full address of the customer, quantity, value, tax, dates of actual supply and preferable proof of satisfactory Performance including after sales service and maintenance support.
5	<i>Bidder (sole)/All Consortium Partners</i> Unconditional acceptance of all commercial terms and conditions.	Self-certificated document to be submitted

BILL OF QUANTITIES									
SL NO	ANNEXURE	DSR 2019	Description	HSN Code	DSR 2019	Sub-description	Unit	Qty.	
			PIPES AND ACCESSORIES						
ITEM NO 1	FIRE FIGHTING WET RISER HYDRANT & SPRINKLE SYSTEM [ANNEXURE-I]	6	Supply of 'C' class heavy duty MS Pipe conforming to IS 1239/3589 i/c fittings like elbows, tees, flanges, tapers, nuts bolts, gaskets etc. in ground including welding, excavation & providing cement concrete blocks as supports, anticorrosive treatment with coaltar/asphalt tape as per IS 10221, refilling the trench etc. of following sizes complete as required. (MAKE: JINDAL (HISSAR), TATA, PRAKASH)	7306	6.2	150 mm. Dia	Metre	78	
ITEM NO 2		7	Supply of 'C' class heavy duty MS pipe conforming to IS 3589/IS 1239 including Welding, fittings like elbows, tees, flanges, tapers, nuts bolts, gaskets etc. and fixing the pipe on the wall/ceiling with suitable clamp/support frame and painting with two or more coats of synthetic enamel paint of required shade complete as required : (MAKE: JINDAL (HISSAR), TATA, PRAKASH)	7306	7.1	25 mm dia	RM	210	
ITEM NO 3					7.2	32 mm dia	RM	294	
ITEM NO 4					7.3	40 mm dia	RM	246	
ITEM NO 5					7.4	50 mm dia	RM	12	
ITEM NO 6					7.5	65 mm dia	RM	12	
ITEM NO 7					7.6	80 mm dia	RM	24	
ITEM NO 8					7.7	100 mm dia	RM	48	
ITEM NO 9					7.8	150 mm dia	RM	126	
ITEM NO 10		9	Supply Single headed external Internal hydrant valve with 1 No. 63 mm dia instantaneous FM Gunmetal/Stainless Steel coupling and cast iron wheel, ISI marked, conforming to IS 5290 (type A) with blank Gunmetal/Stainless Steel cap and chain as required : (MAKE: SAVEMAX, NEWAGE,OMEX)	8481	9.2	Single headed Stainless steel	RM	6	
ITEM NO 11		10	Supply single headed external yard hydrant valve with 1 No. 63 mm dia instantaneous FM Gunmetal/Stainless Steel coupling and cast iron wheel, ISI marked, conforming to IS 5290 (type A) with blank Gunmetal/Stainless Steel cap and chain as required :(MAKE: SAVEMAX, NEWAGE, OMEX)	8481	10.2	Single headed Stainless steel	RM	4	
ITEM NO 12		11	Supply of butterfly valve of PN 1.6 rating with bronze/gunmetal seat duly ISI marked complete with nuts, bolts, washers, gaskets conforming to IS 13095 of following sizes as required :	8481	11.4	80 mm dia	Set	10	
ITEM NO 13					11.5	100 mm dia	Set	2	
ITEM NO 14					11.6	150 mm dia	Set	3	
ITEM NO 15		13	Supply of orifice plate made out of 6 mm thick stainless steel (Grade 304) with orifice of required size to be fitted between flange & landing valve of external and internal hydrants to reduce pressure at the outlet to the level of 3.5 kg/cm2 complete as required. (MAKE: SAVEMAX, NEWAGE, OMEX)	7307			Each	6	
ITEM NO 16		16	Supply 63 mm dia, 15 m long RRL hose pipe with 63 mm dia male and female couplings duly bound with GI wire, rivets etc. conforming to IS 636 (type-A) as required:(MAKE: SAVEMAX/OMEX/NEWAGE)	5909	16.2	Stainless Steel (Grade 304)	Set	20	
ITEM NO 17		17	Supply first-aid Hose Reel with MS construction spray painted in post office red, conforming to IS 884 complete with the following as required. 20 mm nominal internal dia water hose thermoplastic (Textile reinforced) type -2 as per IS: 12585 20 mm nominal internal dia gun metal globe valve & nozzle. Drum and brackets for fixing the equipments on wall. Connections from riser with 25 mm dia stop gun metal valve & M.S. Pipe and socket.(MAKE: SAVEMAX/OMEX/ NEWAGE)	8424	17.1	30 m	Set	4	
ITEM NO 18		21	Supply 15mm dia quartzoid bulb type sprinklers of rating 68 degree centigrade with required accessories :	8424	21.1	Pendent Sprinkler	Each	40	
ITEM NO 19					21.2	Upright Sprinkler	Each	220	
ITEM NO 20		24	Supply installation control valve of cast iron body, brass/bronze working parts comprising of water motor alarm, bronze seat clapper, clapper arm and hydraulically driven mechanical gong bell to sound continuous alarm when the wet riser/sprinkler system activates, pressure gauges, emergency releases, strainer, pressure switch, cock valve complete with drain valve and bypass, test control box, ball valves, MS pipe of required size, flanges, orifice plate, gasket etc of following sizes as required :	8481	24.3	150mm dia	Set	1	
ITEM NO 21		25	Supply sprinkler flexible pipe (UL Listed) of stainless steel complete with 15 NPT on reducer thread with maximum working pressure of 175 PSI test pressure of 875 PSI (Burst) with branch line (Inlet) 25mm NPT male thread to sprinkler head (Outlet) 15mm NPT female thread with reducer, nipple, 2 side brackets, centre bracket, stockbar of following sizes complete as required.	8307	25.3	1200mm	Set	40	
ITEM NO 22		29	Supply adustable rosette plate for 15mm dia in white finish UL Listed or FM approved complete as required.	7907			Each	40	
ITEM NO 23		1	Supply gun metal ball valve suitable for pressure 15 Kg/Cm2 and confirming to IS: 778 of the following size including providing necessary union / flange.	8481	a)	25 mm dia	NOS	4	
ITEM NO 24		5	Supply weather proof cabinet of size not less than 0.75 x 0.6 x 0.25 mtr made out of M.S. sheet not less than 1.5 mm thick having central opening and 4 mm thick glazed glass doors (Two nos.) suitably marked on the outside 'with the letters "FIRE HOSE" including necessary locking arrangement and painting in red colour suitable to accommodate, external yard hydrant valve, 2 nos 15 mtr long Hose pipe, branch pipe, nozzle and fire man's axe. It shall be mounted on boundary wall complete as required.	7310			NOS	6	
			DSR items						
ITEM NO 25	INTELLIGENT ADDRESSABLE FIRE DETECTION & ALARM [ANNEXURE-II]	17.2.4	Supply intelligent analog addressable photothermal detector complete with mounting base complete as required.	8531			Each	12	
ITEM NO 26		17.2.14	Supply addressable manual call point complete as required.	8531			Each	4	
ITEM NO 27		17.2.15	Supply addressable horn cum strobe complete as required.	8531			Each	4	
ITEM NO 28		17.5.2	Supply 2x1.5 sqmm Fire alarm armoured cable, 600/1000V rated with annealed copper conductor having XLPE insulation, steel wire armouring & FRLS outer sheath complete as required.	8544			Metre	800	

		Emergency Exit Light					
ITEM NO 29	EMERGENCY EXIT SIGNAGE [ANNEXURE-III]	Supply wall mounting / ceiling hung type, self contained Rechargeable LED Emergency Egress Routes maintained type Signage/ Escape Route/ Directional Light with LED lamp having multi LED illumination on normal as well as mains failure on self contained battery source complete with G.I. chain, brackets, hardwares etc. as required for fixing. ON resumption of mains supply, the lamp switches back to the mains automatically and simultaneously starts recharging the battery to the required level. The Signage Fitting shall be fabricated from 22SWG C.R.C.A.steel, Epoxy polyester powder coating, White in colour, 2 m.m. thick milky acrylic with vinyl pasted on it, complete with rechargeable Ni-Cd / Ni-Mh battery for 3 Hours backup, 16 to 24 hours recharging period with constant current charger. Input - 230V, 50 HZ, confirms to IS: 10322. (Ref PROLITE Autoglo Model -PEL LED Wall Mounted A.D. or equivalent).	9405			Each	4
ITEM NO 30		Supply wall mounting / ceiling hung type, self contained Rechargeable LED Emergency Egress Routes maintained type Signage/ Escape Route/ Directional Light with LED lamp having multi LED illumination on normal as well as mains failure on self contained battery source complete with G.I. chain, brackets, hardwares etc. as required for fixing. ON resumption of mains supply, the lamp switches back to the mains automatically and simultaneously starts recharging the battery to the required level. The Signage Fitting shall be fabricated from 22SWG C.R.C.A.steel, Epoxy polyester powder coating, White in colour, 2 m.m. thick milky acrylic with vinyl pasted on it, complete with rechargeable Ni-Cd / Ni-Mh battery for 3 Hours backup, 16 to 24 hours recharging period with constant current charger. Input - 230V, 50 HZ, confirms to IS: 10322. (Ref PROLITE Autoglo Model -PEL LED Wall Mounted A.D. or equivalent).	9405			Each	2
ITEM NO 31	SERVICE	Installation, Commissioning & Testing of the complete firefighting Systems				Lum	1

Bidder Security Declaration

Date:

From:

To: Mishra Dhatu Nigam Limited

Tender Reference Number:

Bid No. (If any):

We, the undersigned, declare that: We understand that, according to Midhani tender terms and conditions, bids must be supported by a Bid-Security Declaration. We accept that we will automatically be suspended from being eligible for bidding in any contract with Midhani for the period of time of 2 years without prejudice to Midhani right to take any other action as deemed fit in case we withdraw or change /amend our Bid during the period of bid validity .

We understand this Bid-Securing Declaration shall expire if we are not the successful Bidder.

Signature and seal:

Name:

In the capacity of:

Date:

Note: In case of a Joint Venture, the Bid-Securing Declaration must be signed by all partners to the Joint Venture that submits the bid.

**DECLARATION TO BE GIVEN BY VENDORS ALONG WITH BID (MANDATORY) ON THEIR LETTER HEAD
IN CASE THE VALUE OF PROCUREMENT IS MORE THAN 5 LAKHS**

Tender Ref No: With reference to the above mentioned RFP/Tender, we hereby confirm that we have read the Public Procurement (Preference to Make in India) Order No. P-45021/2/2017-PP (BE-II) revision dated 16/9/2020 and the updates and the percentage of local (Indian) content in the offered items/service is _____ and the place/places of value addition done is/are _____.

Signature with seal

Note: In cases of procurement for value in excess of 10 Crores the Class-I /Class-II local supplier shall be required to provide a certificate from statutory auditor or cost auditor (In case of companies) or from a practicing cost accountant or practicing chartered accountant (In respect of suppliers other than companies) giving the percentage of local content.

GENERAL TERMS AND CONDITIONS FOR INDEGENOUS SOURCES**1. DEFINITIONS:****PURCHASER:**

The term "Purchaser" as used herein shall mean Mishra Dhatu Nigam Limited, incorporated under the Companies Act, 1956, and having its registered office at P.O. Kanchanbagh, Hyderabad - 500 058, India, and shall include its successors and assigns.

TENDERER:

The term "Tenderer" shall mean the person, firm or corporation submitting a tender against the Invitation to tender and shall include his/its heirs, executors, administrators, legal representatives, successors and assigns.

SUCCESSFUL TENDERER/SUPPLIER/CONTRACTOR:

The term "Successful Tenderer/Supplier/Contractor" shall mean the Tenderer whose tender has been accepted and shall include his/its heirs, executors, administrators, legal representatives, successors and assigns approved by the Purchaser.

2. The Tenderer shall indicate/furnish the following in his offer (if applicable):

- a. Manufacturer's Name & Address.
- b. Country of origin of goods.
- c. Delivery period.
- d. Mode of packing.
- e. Net weight and Gross weight.
- f. Technical Literature / Catalogue/Specifications.
- g. Details of Shelf life, if any, applicable for the goods offered.
- h. Your Banker's Name, Address & Details.
- i. A confirmation that the price quoted is your lowest export price and is the same as you would normally quote to Government department and other favored customers.
- j. HSN Code for all the Items quoted along with applicable GST Rates.
- k. TARIFF HEAD UNDER H. S. (HARMONISED SYSTEM).

3. Any Order resulting from this invitation to tender shall be governed by our General Terms and Conditions of Contract and the supplier quoting against this enquiry Shall be deemed / to have read and understood the same.**4. Where counter terms and conditions have been offered by the Tenderer, the purchaser shall not be governed by these unless specific acceptances have been given in writing in the order by the Purchaser****5. The offer should be complete in all respects. Full particulars and descriptive literature and drawings should be forwarded along with the quotation. The make of the items offered should be clearly specified. Materials should be offered strictly confirming to our specifications. The deviations if any, should be clearly indicated in the quotation. Test certificates must be produced, wherever required where stipulated samples should be forwarded along with the quotation. Material confirming to IS will be preferred.****6. PRICES:**

Price Basis: FOR Midhani, Hyderabad basis shall be quoted. The prices must be unit shown inclusive of packing, forwarding, insurance, Octroi duty and delivery charges and should be on FOR destination basis. Offers from local suppliers should be for free delivery at our stores

GENERAL TERMS AND CONDITIONS FOR INDEGENOUS SOURCES**7. TAXES:**

The Tenderer shall indicate their GST Number & date in the quotation. Taxes payable have to be clearly indicated in the quotation. The nature of duties and applicable percentages thereof and the items on which such taxes are leviable along with applicable HSN code shall be indicated in the Offer. Payment of taxes shall be as applicable at the time of supply and on production of documentary evidence. Statutory variation in the rate of Taxes during the delivery period shall be accepted.

8. RIGHT OF ACCEPTANCE:

The purchaser does not bind himself to accept the lowest or any other tender and reserves the right of accepting the whole or any part of the tender or portion of the quantity offered. Within a period of six months from the date of completion of the order, the Purchaser further reserves the right to place order on the successful tenderer for an additional upto 100% of the original ordered quantity.

9. INSPECTION:

The Purchaser reserves the right to get the Stores inspected before commencement of supplier/ or on arrival at our factory by an inspecting officer authorized by the Purchaser. The Tenderer should be in a position show and/ or send samples of material on demand.

10. PACKING:

The stores should be securely packed and properly marked to avoid loss or damage in Transit by Rail /Road.

11. TO RECOVER LIQUIDATED DAMAGES:

Liquidated Damages shall be levied against Suppliers/Contractors in the event of unsatisfactory, delayed or non supply of materials/execution of Contract beyond the date of delivery/completion of job. LD is leviable at the rate of 1% of the total order/contract prices per week or part there of subject to a maximum 10% of the order/contract prices including taxes & duties without prejudice to the right of the purchaser to take any other action. The amount shall also be recoverable from any other contract on account of the supplier. The payment or deduction of such damages shall not relieve the contractor from his obligations to complete the work or from any other of his obligations and liabilities under the contract. MIDHANI shall issue GST Invoice/Debit Note for the LD deducted on the name of Supplier/Contractor against which GST credit may be availed, if applicable

12. RISK PURCHASE:

The supply of all items must be completed satisfactory and within the specified period in the order failing which, the Purchaser reserves the right to purchase stores from other sources at the suppliers cost and risk.

13. GUARANTEE:

The supplier shall guarantee the store supplied in respect of good design, quality and workmanships for a period of 12 months from the commissioning or 18 months from the date of supply whichever is earlier.

14. VALIDITY:

The offer should be valid for a minimum period of 180 days from the date of opening of tender.

GENERAL TERMS AND CONDITIONS FOR INDEGENOUS SOURCES

15. ADDENDA TO TENDER DOCUMENTS:

The Purchaser reserves the right to issue addenda to the tender documents to clarify, modify, supplement or delete any of the condition, clause or items stated in the tender documents issued with this invitation to Tender. Each addendum issued will be distributed to each tenderer or his authorized representative and the addendum so issued shall form a part of the original tender documents to be reviewed as required.

16. In case the item offered is covered by DGS & D Rae Contract, the rate should be as per R/C and a copy of R/C should be enclosed with the offer.

17. EARNEST MONEY DEPOSIT (EMD):

Tenderer/Bidder are required to furnish EMD of amount specified in tender (NIT) Through Online or in the form Bank Guarantee (BG) is acceptable.

All Bidders are requested to Submit the "EMD & Security Deposit "online through the below Link: <http://ebs.in/midhani/public/> (or)

Visit Midhani website <http://www.midhani-india.in> > Purchase > Tenders > **TENDER FEE, EMD FEE(EARNEST MONEY DEPOSIT) AND SECURITY DEPOSIT - ONLINE PAYMENT**

In case of EMD/SD in the form of Bank Guarantee(BG), bidder shall submit BG as per format in the tender document,

- i. In Indian Currency – from a Nationalized Bank or Scheduled Bank encashable in India.
- ii. In Foreign Currency – from SBI, Frankfurt/ SBI, New York / any Nationalized Bank or Scheduled Bank encashable in India.
- iii. EMD Bank guarantee shall be valid for a period of 90 days.
- iv. Security Deposit Bank guarantee shall be valid till material is received and accepted at Midhani.

EMD Exemption:

(I) As per Public Procurement Policy(PPP) for Micro & Small Enterprises (MSEs) order, 2012 vide Gazette notification dated 23.03.2012 by Ministry of Micro, Small and Medium Enterprises of Govt of India, the following benefits will be extended to the MSEs.

EMD is exempted for Micro & Small Enterprises(MSEs) registered with DIC/NSIC/KVIC/Udyog Aadhar Memorandum (UAM) issued by MoMSME or any other body specified by Ministry of MSME.

Purchase Preference: The following Purchase preference procedure is followed as per Ministry guidelines for award of quantity/order

Type of Tender	Price quoted by MSE	Finalization of tender
(i) If tender can be split	L-1	Full order on MSE
(ii) If tender can be Split	Not L-1 but within L1+15%	20% order on MSE subject to matching with L-1 price
(iii) If tender cannot be split	L-1	Full order on MSE
(iv) If tender cannot be split	Not L-1 but within L-1 +15%	Full order on MSE subject to matching with L-1 Price

GENERAL TERMS AND CONDITIONS FOR INDEGENOUS SOURCES

Note:

- (i) In case more than one MSE enterprise quotes same price, the supply/order shall be shared proportionally (to tender quantity) if tender can be split.
- (ii) If the MSE who have quoted lowest rate among the MSEs in the price band of L-1 + 15% do not agree to match the rate of L-1 of the tender, then the next ranked MSE bidder who has quoted within the price band of L1+15% in order shall be given chance to match the rate of L-1 for award of the quantity/order.
- (ii) All Central PSU's & Government departments / Authorities as well as Original Raw Material Manufacturers may be exempt from submission of EMD in all tenders.

18. SECURITY DEPOSIT (SD):

The successful tenderer/Bidder shall be required to furnish SD as specified in tender (NIT) within 21 days from the date of Purchase Order. In case Security Deposit is not submitted within the stipulated time as above, interest @ 12% p.a. shall be levied for the period of delay beyond the stipulated time. Interest as above may be either deposited by the supplier / contractor or recovered from any amounts due to the supplier / contractor.

The security deposit shall be for the due and faithful performance of the contract and shall remain binding notwithstanding such variations, alterations or extensions of time as it may be made, given, conceded or agreed to between the Supplier/Contractor and Purchaser.

The Security Deposit furnished by the successful tenderer will be subject to the Terms & Conditions of the order/contract finally concluded between the parties and the Purchaser will not be liable for payment of any interest on the security deposit or any depreciation thereof.

The Security Deposit shall be refunded on application by the contractor after expiry of the contract period and after he has discharged all his obligations under the contract and produced a certificate from the Purchaser's authorized representatives certifying the due completion & acceptance of the work.

All Government Departments and Central PSUs are exempted from payment of Security Deposit. In all cases where SD is exempted, in case of failure of the bidder to accept / execute the contract as per agreed terms, the bidder shall not be permitted to participate in the re-tender for the same item. Suitable penal action in accordance with other provisions of the Tender shall also be applicable.

19. MAKE IN INDIA:

Public Procurement (Preference to Make in India) Clause:

Purchase preference will be granted as per Public Procurement (Preference to Make in India), Order-2017 as amended, issued by DPIIT/Ministry of Commerce and Industry.

This tender notice/RFP complies with Public Procurement (Preference to Make in India), Order 2017 by Department of Industrial Policy and Promotion, Ministry of Commerce and Industry Government of India issued vide letter No. P-45021/2/2017-B.E-II dated: 15th June'2017 and as amended. (For latest updates refer to www.dpiit.gov.in). The following is add-on information. However it is mandatory to refer website www.dpiit.gov.in for latest update prior to responding for this tender. Brief of recent amendment issued vide letter No: P-45021/2/2017-PP (BE-II) dated: 16th Sept'2020 is given below.

GENERAL TERMS AND CONDITIONS FOR INDEGENOUS SOURCES**(a) Types of Suppliers:**

- Class – I Local Supplier : Local content = 50% Minimum
- Class – II Local Supplier : Local content = 20% Minimum and less than 50%
- Non-Local Supplier : Local content = Less than 20%

(b) Margin of purchase preference = 20%**(c) Preference (s):**

- i. “Class-I Local Supplier” is only eligible to bid irrespective of bid value for which Nodal ministry has declared there is sufficient local capacity available for the tendered stores/service.
- ii. “Class-I Local” Supplier and “Class-II Local Supplier” as defined under the order shall be eligible to bid for procurements except when Global tender enquiry has been issued. In Global tender enquiries, non local suppliers shall also be eligible to bid along with ‘Class-I Local Supplier’ and “Class-II Local Suppliers’.
- iii. Works includes Engineering, procurement and construction (EPC) contracts and services include system integrator (SI) contracts

(d) L1-Declaration Methodology for divisible goods or works:

- i. Among all qualified bids, Lowest bid is termed as L1
- ii. If L1 = Class-I Local Supplier, full quantity will be awarded to L1;
- iii. If L1 bid is not Class-I local supplier, 50% of the order quantity shall be awarded to L1 and balance 50% quantity will be awarded to the lowest bidder among the “Class-I Local Suppliers” to match the L1 price and subject to his quoted price falling within 20% of L1 price; If fails to match, the same will be offered to next higher bidder among Class-I Local Suppliers and so on. In case some quantity left uncovered on class-I local suppliers, such balance quantity shall be awarded to L1.

(e) L1-Declaration Methodology for non-divisible goods or works:

- i. Among all qualified bids, Lowest bid is termed as L1
- ii. If L1 = Class-I Local Supplier, order will be awarded to L1
- iii. If L1 = Not a Class-I Local Supplier then order will be awarded to the lowest bidder among the “Class-I Local Suppliers” to match the L1 price subject to his quoted price falling within 20% of L1 price.
- iv. If fails to match, the same will be offered to next higher bidder among Class-I Local Suppliers and so on.
- v. In case none of the Class-I Local Suppliers matches the L1 Price, then order shall be awarded to L1.

(f) Verification of Local Content:

- i. The Class-I Local Supplier / Class-II Local Supplier need to provide percentage of local content, location(s) of value addition and self-certification to this effect along with their technical bidding document.
- ii. For procurement value more than Rs. 10 Crores, the Class -I Local Supplier / Class-II Local Supplier need to provide percentage of local content, location(s) of value addition and certification to this effect obtained from Statutory Auditor / Cost Auditor (for companies) or from Practicing Cost Accountant or Practicing Chartered Accountant (for other than companies) along with their technical bidding document.
- iii. In case any Bidder provides false certification with respect to local contents etc, bidder or their successor(s) can be debarred up to two years as per

GENERAL TERMS AND CONDITIONS FOR INDEGENOUS SOURCES

provisions of GFR along with such other actions as may be permissible under law.

Note: For complete details, please refer Public Procurement (Preference to Make in India) Order 2017 P-45021/2/217-PP (BE-II) revision dated: 16th Sept'2020 (Enclosed)

20. NO CLAIM OR COMPENSATION FOR SUBMISSION OF TENDER:

The tenderers whose tenders are not accepted shall not be entitled to claim any costs, charges and expenses of the tender, incidental to or incurred by them, through or in connection with their submission of tenders even though the Purchaser may elect to withdraw the invitation to tender.

21. LOWEST TENDER NOT NECESSARY TO BE ACCEPTED:

The Purchaser is not bound to accept the lowest or any tender or to assign reasons for its non-acceptance. The Purchaser also reserves the right to accept the tender either in whole or in part.

22. PRICE NEGOTIATIONS:

Price negotiations as such shall not be held, except in the case of negotiations with the lowest tenderer and accordingly, the tenderers shall have to submit their best commercial bids.

23. LOADING IN CASE OF COMMERCIAL DEVIATIONS:

In case of any acceptable commercial deviation, MIDHANI may evaluate the prices with appropriate loading at One year MCLR rate + 0.5% of SBI prevailing on the date of Technical bid opening.

The comparative assessment of offers received would be made on equal footing taking into account the financial implications for the deviations in terms and conditions. Conditional discounts offered by the tenderers for coverage within a shorter period for early inspection / payment etc., shall not be considered at the time of evaluation of tenders.

24. ARBITRATION:

Any dispute(s) or difference(s) whatsoever arises under or out of or in connection with the EOI/contract, or in respect of any defined legal relationship associated therewith or derived there from, shall be resolved/settled amicably, through mutual negotiation; failing which the differences shall be resolved by way of arbitration in accordance with the International Centre for Alternative Dispute Resolution (ICADR) Arbitration Rules 1996. The authority to appoint the arbitrator(s) shall be the International Centre for Alternative Dispute Resolution (ICADR). And will provide administrative services in accordance with ICADR Arbitration Rules 1996. The seat of arbitration shall be India. The language of the arbitration proceeding shall be English. The place of arbitration proceedings shall be Hyderabad, Telangana, India. In case of PSU/Government organization DPE guidelines in force or as amended from time to time shall be applicable. In the event of any dispute or difference relating to the interpretation and application of the provisions of commercial contract(s) between Central Public Sector Enterprises (CPSEs)/Port Trusts inter se and also between CPSEs and Government dept./Organizations (Excluding disputes concerning Railways, Income Tax, Customs & Excise Dept.), such disputes or differences shall be taken up by either party for resolution through AMRCD as mentioned in DPE OM No. 4(1)/ 2013- DPE(GM)/FTS-1835 dated 22/5/2018 or the latest DPE Guidelines issued from time to time.

GENERAL TERMS AND CONDITIONS FOR INDEGENOUS SOURCES**25. JURISDICTION:**

All questions, disputes or differences arising under, out of or in connection with the contract shall be subject to the exclusive jurisdiction of court within local limits of Hyderabad, India.

26. BANKRUPTCY ETC.:

If the Supplier/Contractor shall become bankrupt or insolvent or cause or suffer any receiver to be appointed of his business or any asset thereof or compound with his creditors, or being a corporation commence to be wound up, or carry on its business under a Receiver for the benefits of its creditors or any of them, the Purchaser shall be at liberty:

- a. To terminate the contract forthwith upon coming to know of the happening of any such event as aforesaid by notice in writing to the Contractor or to the receiver / liquidator.
- OR
- b. To give such receiver, liquidator or other person the option of carrying out the contract subject to his providing guarantee up to an amount to be agreed for the due and faithful performance of the Contract.

27. CANCELLATION / SHORT CLOSURE OF PURCHASE ORDER:

The purchaser may, without prejudice to any other remedy for breach of Contract, by written notice of default sent to the supplier, terminate the contract in whole or in part:

- a) If the supplier fails to deliver any or all of the stores within the time period(s) specified in the contract, or any extension thereof granted by the Purchaser.
- b) If the supplier fails to perform any other obligation under the contract within the period specified in the contract or any extension thereof granted by the purchaser.
- c) Purchaser reserves the right to cancel the Purchase Order/ contract on its own under exceptional circumstances.

On receipt of notice for short closure, the contractor shall cease all further work, except for such work as may be specified in the notice for the sole purpose of protecting that part of the stores already executed. Further, the liability of MIDHANI in such cases will be limited to the extent of the cost as assessed by MIDHANI, in its opinion, till the point of short closure

28. REJECTED GOODS:

The rejected goods at MIDHANI premises will lie at the risk of the Vendor. They should be replaced immediately on receipt of intimation regarding rejection from Midhani. In case the rejected goods are not replaced within six weeks from the date of receipt of the inspection memo, MIDHANI is not responsible for any loss, damage and theft, and in addition, MIDHANI reserves the right to recover such amount as may be assessed on account of storage and incidentals or any other charges MIDHANI may have incurred.

The vendor has to lift the rejected material from MIDHANI within 6 months from the date of receipt of intimation. In case of rejected goods are in MIDHANI for more than 6 months, MIDHANI reserves the right to dispose of the same and Vendor shall not have any right for claiming the goods or damages thereof.

29. Only one vertical (Company) from a group of companies will be eligible to participate in the tender. MIDHANI reserves the right to reject the bids of all such parties summarily, if it is found that verticals under the same organizations have participated.

30. COO (Country Of Origin) certificate to be issued by chambers of commerce to avail discount in preferential custom duty amount in Indian customs.

GENERAL TERMS AND CONDITIONS FOR INDEGENOUS SOURCES**31. COUNTERFEIT PARTS:**

MIDHANI requires its supplier to take steps to eliminate the counterfeit (or) suspect counterfeit materials/ parts/ components. We define Counterfeit work as items that are, or contain, unlawful or unauthorized reproductions, substitutions or alterations that have been knowingly mismarked, misidentified or otherwise misrepresented to be an authentic, unmodified material/ parts/ component from the original manufacturer. Unlawful or unauthorized substitution includes used items represented as new or the false identification of Grade, Serial number / Heat number, Lot number and date code or performance characteristics/Test certificate/Coc / CoA. Suppliers of raw materials/ parts/ components are used fulfilling Purchase orders of MIDHANI LTD. Distributors shall only purchase materials/ components directly from original manufacturers.

32. TRADE PRACTICES:

Supplier will, at all times, refrain from engaging in any illegal, unfair or deceptive trade practices or unethical business practices. Supplier shall at its expense obtain any and all permits, licences, authorizations, and/or certificates that may be required in any jurisdiction or by any regulatory or administrative agency in connection with its activities hereunder. Supplier shall ensure that its employees are aware of:

- Their contribution to product or service conformity
- Their contribution to product safety, and
- Their contribution to the importance of ethical behaviour.

No. P-45021/2/2017-PP (BE-II)
Government of India
Ministry of Commerce and Industry
Department for Promotion of Industry and Internal Trade
(Public Procurement Section)

Udyog Bhawan, New Delhi
Dated: 16th September, 2020

To

All Central Ministries/Departments/CPSUs/All concerned

ORDER

Subject: Public Procurement (Preference to Make in India), Order 2017– Revision; regarding.

Department for Promotion of Industry and Internal Trade, in partial modification [Paras 2, 3, 5, 10 & 13] of Order No.P-45021/2/2017-B.E.-II dated 15.6.2017 as amended by Order No.P-45021/2/2017-B.E.-II dated 28.05.2018, Order No.P-45021/2/2017-B.E.-II dated 29.05.2019 and Order No.P-45021/2/2017-B.E.-II dated 04.06.2020, hereby issues the revised 'Public Procurement (Preference to Make in India), Order 2017" dated 16.09.2020 effective with immediate effect.

Whereas it is the policy of the Government of India to encourage 'Make in India' and promote manufacturing and production of goods and services in India with a view to enhancing income and employment, and

Whereas procurement by the Government is substantial in amount and can contribute towards this policy objective, and

Whereas local content can be increased through partnerships, cooperation with local companies, establishing production units in India or Joint Ventures (JV) with Indian suppliers, increasing the participation of local employees in services and training them,

Now therefore the following Order is issued:

1. This Order is issued pursuant to Rule 153 (iii) of the General Financial Rules 2017.
2. **Definitions:** For the purposes of this Order:

'Local content' means the amount of value added in India which shall, unless otherwise prescribed by the Nodal Ministry, be the total value of the item procured (excluding net domestic indirect taxes) minus the value of imported content in the item (including all customs duties) as a proportion of the total value, in percent.

'Class-I local supplier' means a supplier or service provider, whose goods, services or works offered for procurement, meets the minimum local content as prescribed for 'Class-I local supplier' under this Order.

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'Class-II local supplier' means a supplier or service provider, whose goods, services or works offered for procurement, meets the minimum local content as prescribed for *'Class-II local supplier'* but less than that prescribed for *'Class-I local supplier'* under this Order.

'Non - Local supplier' means a supplier or service provider, whose goods, services or works offered for procurement, has local content less than that prescribed for *'Class-II local supplier'* under this Order.

'L1' means the lowest tender or lowest bid or the lowest quotation received in a tender, bidding process or other procurement solicitation as adjudged in the evaluation process as per the tender or other procurement solicitation.

'Margin of purchase preference' means the maximum extent to which the price quoted by a *"Class-I local supplier"* may be above the L1 for the purpose of purchase preference.

'Nodal Ministry' means the Ministry or Department identified pursuant to this order in respect of a particular item of goods or services or works.

'Procuring entity' means a Ministry or department or attached or subordinate office of, or autonomous body controlled by, the Government of India and includes Government companies as defined in the Companies Act.

'Works' means all works as per Rule 130 of GFR- 2017, and will also include *'turnkey works'*.

3. Eligibility of *'Class-I local supplier'*/ *'Class-II local supplier'*/ *'Non-local suppliers'* for different types of procurement

(a) In procurement of all goods, services or works in respect of which the Nodal Ministry / Department has communicated that there is sufficient local capacity and local competition, only *'Class-I local supplier'*, as defined under the Order, shall be eligible to bid irrespective of purchase value.

(b) Only *'Class-I local supplier'* and *'Class-II local supplier'*, as defined under the Order, shall be eligible to bid in procurements undertaken by procuring entities, except when Global tender enquiry has been issued. In global tender enquiries, *'Non-local suppliers'* shall also be eligible to bid along with *'Class-I local suppliers'* and *'Class-II local suppliers'*. In procurement of all goods, services or works, not covered by sub-para 3(a) above, and with estimated value of purchases less than Rs. 200 Crore, in accordance with Rule 161(iv) of GFR, 2017, Global tender enquiry shall not be issued except with the approval of competent authority as designated by Department of Expenditure.

(c) For the purpose of this Order, works includes Engineering, Procurement and Construction (EPC) contracts and services include System Integrator (SI) contracts.

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3A. Purchase Preference

(a) Subject to the provisions of this Order and to any specific instructions issued by the Nodal Ministry or in pursuance of this Order, purchase preference shall be given to 'Class-I local supplier' in procurements undertaken by procuring entities in the manner specified here under.

(b) In the procurements of goods or works, which are covered by para 3(b) above and which are divisible in nature, the 'Class-I local supplier' shall get purchase preference over 'Class-II local supplier' as well as 'Non-local supplier', as per following procedure:

- i. Among all qualified bids, the lowest bid will be termed as L1. If L1 is 'Class-I local supplier', the contract for full quantity will be awarded to L1.
- ii. If L1 bid is not a 'Class-I local supplier', 50% of the order quantity shall be awarded to L1. Thereafter, the lowest bidder among the 'Class-I local supplier' will be invited to match the L1 price for the remaining 50% quantity subject to the Class-I local supplier's quoted price falling within the margin of purchase preference, and contract for that quantity shall be awarded to such 'Class-I local supplier' subject to matching the L1 price. In case such lowest eligible 'Class-I local supplier' fails to match the L1 price or accepts less than the offered quantity, the next higher 'Class-I local supplier' within the margin of purchase preference shall be invited to match the L1 price for remaining quantity and so on, and contract shall be awarded accordingly. In case some quantity is still left uncovered on Class-I local suppliers, then such balance quantity may also be ordered on the L1 bidder.

(c) In the procurements of goods or works, which are covered by para 3(b) above and which are not divisible in nature, and in procurement of services where the bid is evaluated on price alone, the 'Class-I local supplier' shall get purchase preference over 'Class-II local supplier' as well as 'Non-local supplier', as per following procedure:

- i. Among all qualified bids, the lowest bid will be termed as L1. If L1 is 'Class-I local supplier', the contract will be awarded to L1.
- ii. If L1 is not 'Class-I local supplier', the lowest bidder among the 'Class-I local supplier', will be invited to match the L1 price subject to Class-I local supplier's quoted price falling within the margin of purchase preference, and the contract shall be awarded to such 'Class-I local supplier' subject to matching the L1 price.
- iii. In case such lowest eligible 'Class-I local supplier' fails to match the L1 price, the 'Class-I local supplier' with the next higher bid within the margin of purchase preference shall be invited to match the L1 price and so on and contract shall be awarded accordingly. In case none of the 'Class-I local supplier' within the margin of purchase preference matches the L1 price, the contract may be awarded to the L1 bidder.

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- (d) "Class-II local supplier" will not get purchase preference in any procurement undertaken by procuring entities.

3B. Applicability in tenders where contract is to be awarded to multiple bidders -
In tenders where contract is awarded to multiple bidders subject to matching of L1 rates or otherwise, the 'Class-I local supplier' shall get purchase preference over 'Class-II local supplier' as well as 'Non-local supplier', as per following procedure:

- a) In case there is sufficient local capacity and competition for the item to be procured, as notified by the nodal Ministry, only Class I local suppliers shall be eligible to bid. As such, the multiple suppliers, who would be awarded the contract, should be all and only 'Class I Local suppliers'.
- b) In other cases, 'Class II local suppliers' and 'Non local suppliers' may also participate in the bidding process along with 'Class I Local suppliers' as per provisions of this Order.
- c) If 'Class I Local suppliers' qualify for award of contract for at least 50% of the tendered quantity in any tender, the contract may be awarded to all the qualified bidders as per award criteria stipulated in the bid documents. However, in case 'Class I Local suppliers' do not qualify for award of contract for at least 50% of the tendered quantity, purchase preference should be given to the 'Class I local supplier' over 'Class II local suppliers'/'Non local suppliers' provided that their quoted rate falls within 20% margin of purchase preference of the highest quoted bidder considered for award of contract so as to ensure that the 'Class I Local suppliers' taken in totality are considered for award of contract for at least 50% of the tendered quantity.
- d) First purchase preference has to be given to the lowest quoting 'Class-I local supplier', whose quoted rates fall within 20% margin of purchase preference, subject to its meeting the prescribed criteria for award of contract as also the constraint of maximum quantity that can be sourced from any single supplier. If the lowest quoting 'Class-I local supplier', does not qualify for purchase preference because of aforesaid constraints or does not accept the offered quantity, an opportunity may be given to next higher 'Class-I local supplier', falling within 20% margin of purchase preference, and so on.
- e) To avoid any ambiguity during bid evaluation process, the procuring entities may stipulate its own tender specific criteria for award of contract amongst different bidders including the procedure for purchase preference to 'Class-I local supplier' within the broad policy guidelines stipulated in sub-paras above.

4. **Exemption of small purchases:** Notwithstanding anything contained in paragraph 3, procurements where the estimated value to be procured is less than Rs. 5 lakhs shall be exempt from this Order. However, it shall be ensured by procuring entities that procurement is not split for the purpose of avoiding the provisions of this Order.
5. **Minimum local content:** The 'local content' requirement to categorize a supplier as 'Class-I local supplier' is minimum 50%. For 'Class-II local supplier', the 'local content' requirement is minimum 20%. Nodal Ministry/ Department may prescribe only a higher

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percentage of minimum local content requirement to categorize a supplier as 'Class-I local supplier'/'Class-II local supplier'. For the items, for which Nodal Ministry/ Department has not prescribed higher minimum local content notification under the Order, it shall be 50% and 20% for 'Class-I local supplier'/'Class-II local supplier' respectively.

6. **Margin of Purchase Preference:** The margin of purchase preference shall be 20%.
7. **Requirement for specification in advance:** The minimum local content, the margin of purchase preference and the procedure for preference to Make in India shall be specified in the notice inviting tenders or other form of procurement solicitation and shall not be varied during a particular procurement transaction.
8. **Government E-marketplace:** In respect of procurement through the Government E-marketplace (GeM) shall, as far as possible, specifically mark the items which meet the minimum local content while registering the item for display, and shall, wherever feasible, make provision for automated comparison with purchase preference and without purchase preference and for obtaining consent of the local supplier in those cases where purchase preference is to be exercised.
9. **Verification of local content:**
 - a. The 'Class-I local supplier'/'Class-II local supplier' at the time of tender, bidding or solicitation shall be required to indicate percentage of local content and provide self-certification that the item offered meets the local content requirement for 'Class-I local supplier'/'Class-II local supplier', as the case may be. They shall also give details of the location(s) at which the local value addition is made.
 - b. In cases of procurement for a value in excess of Rs. 10 crores, the 'Class-I local supplier'/'Class-II local supplier' shall be required to provide a certificate from the statutory auditor or cost auditor of the company (in the case of companies) or from a practicing cost accountant or practicing chartered accountant (in respect of suppliers other than companies) giving the percentage of local content.
 - c. Decisions on complaints relating to implementation of this Order shall be taken by the competent authority which is empowered to look into procurement-related complaints relating to the procuring entity.
 - d. Nodal Ministries may constitute committees with internal and external experts for independent verification of self-declarations and auditor's/ accountant's certificates on random basis and in the case of complaints.
 - e. Nodal Ministries and procuring entities may prescribe fees for such complaints.
 - f. False declarations will be in breach of the Code of Integrity under Rule 175(1)(i)(h) of the General Financial Rules for which a bidder or its successors can be debarred for up to two years as per Rule 151 (iii) of the General Financial Rules along with such other actions as may be permissible under law.

- g. A supplier who has been debarred by any procuring entity for violation of this Order shall not be eligible for preference under this Order for procurement by any other procuring entity for the duration of the debarment. The debarment for such other procuring entities shall take effect prospectively from the date on which it comes to the notice of other procurement entities, in the manner prescribed under paragraph 9h below.
- h. The Department of Expenditure shall issue suitable instructions for the effective and smooth operation of this process, so that:
 - i. The fact and duration of debarment for violation of this Order by any procuring entity are promptly brought to the notice of the Member-Convener of the Standing Committee and the Department of Expenditure through the concerned Ministry /Department or in some other manner;
 - ii. on a periodical basis such cases are consolidated and a centralized list or decentralized lists of such suppliers with the period of debarment is maintained and displayed on website(s);
 - iii. in respect of procuring entities other than the one which has carried out the debarment, the debarment takes effect prospectively from the date of uploading on the website(s) in the such a manner that ongoing procurements are not disrupted.

10. Specifications in Tenders and other procurement solicitations:

- a. Every procuring entity shall ensure that the eligibility conditions in respect of previous experience fixed in any tender or solicitation do not require proof of supply in other countries or proof of exports.
- b. Procuring entities shall endeavour to see that eligibility conditions, including on matters like turnover, production capability and financial strength do not result in unreasonable exclusion of 'Class-I local supplier' / 'Class-II local supplier' who would otherwise be eligible, beyond what is essential for ensuring quality or creditworthiness of the supplier.
- c. Procuring entities shall, within 2 months of the issue of this Order review all existing eligibility norms and conditions with reference to sub-paragraphs 'a' and 'b' above.
- d. **Reciprocity Clause**
 - i. When a Nodal Ministry/Department identifies that Indian suppliers of an item are not allowed to participate and/ or compete in procurement by any foreign government, due to restrictive tender conditions which have direct or indirect effect of barring Indian companies such as registration in the procuring country, execution of projects of specific value in the procuring country etc., it shall provide such details to all its procuring entities including CMDs/CEOs of PSEs/PSUs, State Governments and other procurement agencies under their administrative control and GeM for appropriate reciprocal action.

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- ii. Entities of countries which have been identified by the nodal Ministry/Department as not allowing Indian companies to participate in their Government procurement for any item related to that nodal Ministry shall not be allowed to participate in Government procurement in India for all items related to that nodal Ministry/ Department, except for the list of items published by the Ministry/ Department permitting their participation.
 - iii. The stipulation in (ii) above shall be part of all tenders invited by the Central Government procuring entities stated in (i) above. All purchases on GeM shall also necessarily have the above provisions for items identified by nodal Ministry/ Department.
 - iv. State Governments should be encouraged to incorporate similar provisions in their respective tenders.
 - v. The term 'entity' of a country shall have the same meaning as under the FDI Policy of DPIIT as amended from time to time.
- e. Specifying foreign certifications/ unreasonable technical specifications/ brands/ models in the bid document is restrictive and discriminatory practice against local suppliers. If foreign certification is required to be stipulated because of non-availability of Indian Standards and/or for any other reason, the same shall be done only after written approval of Secretary of the Department concerned or any other Authority having been designated such power by the Secretary of the Department concerned.
- f. "All administrative Ministries/Departments whose procurement exceeds Rs. 1000 Crore per annum shall notify/ update their procurement projections every year, including those of the PSEs/PSUs, for the next 5 years on their respective website."

10A. Action for non-compliance of the Provisions of the Order: In case restrictive or discriminatory conditions against domestic suppliers are included in bid documents, an inquiry shall be conducted by the Administrative Department undertaking the procurement (including procurement by any entity under its administrative control) to fix responsibility for the same. Thereafter, appropriate action, administrative or otherwise, shall be taken against erring officials of procurement entities under relevant provisions. Intimation on all such actions shall be sent to the Standing Committee.

11. Assessment of supply base by Nodal Ministries: The Nodal Ministry shall keep in view the domestic manufacturing / supply base and assess the available capacity and the extent of local competition while identifying items and prescribing the higher minimum local content or the manner of its calculation, with a view to avoiding cost increase from the operation of this Order.

12. Increase in minimum local content: The Nodal Ministry may annually review the local content requirements with a view to increasing them, subject to availability of sufficient local competition with adequate quality.

13. Manufacture under license/ technology collaboration agreements with phased indigenization: While notifying the minimum local content, Nodal Ministries may make special provisions for exempting suppliers from meeting the stipulated local content if the product is being manufactured in India under a license from a foreign manufacturer who holds intellectual property rights and where there is a technology collaboration agreement / transfer of technology agreement for indigenous manufacture of a product developed abroad with clear phasing of increase in local content.

13A. In procurement of all goods, services or works in respect of which there is substantial quantity of public procurement and for which the nodal ministry has not notified that there is sufficient local capacity and local competition, the concerned nodal ministry shall notify an upper threshold value of procurement beyond which foreign companies shall enter into a joint venture with an Indian company to participate in the tender. Procuring entities, while procuring such items beyond the notified threshold value, shall prescribe in their respective tenders that foreign companies may enter into a joint venture with an Indian company to participate in the tender. The procuring Ministries/Departments shall also make special provisions for exempting such joint ventures from meeting the stipulated minimum local content requirement, which shall be increased in a phased manner.

14. Powers to grant exemption and to reduce minimum local content: The administrative Department undertaking the procurement (including procurement by any entity under its administrative control), with the approval of their Minister-in-charge, may by written order, for reasons to be recorded in writing,

- a. reduce the minimum local content below the prescribed level; or
- b. reduce the margin of purchase preference below 20%; or
- c. exempt any particular item or supplying entities from the operation of this Order or any part of the Order.

A copy of every such order shall be provided to the Standing Committee and concerned Nodal Ministry / Department. The Nodal Ministry / Department concerned will continue to have the power to vary its notification on Minimum Local Content.

15. Directions to Government companies: In respect of Government companies and other procuring entities not governed by the General Financial Rules, the administrative Ministry or Department shall issue policy directions requiring compliance with this Order.

16. Standing Committee: A standing committee is hereby constituted with the following membership:

Secretary, Department for Promotion of Industry and Internal Trade—Chairman
Secretary, Commerce—Member
Secretary, Ministry of Electronics and Information Technology—Member
Joint Secretary (Public Procurement), Department of Expenditure—Member
Joint Secretary (DPIIT)—Member-Convenor

The Secretary of the Department concerned with a particular item shall be a member in respect of issues relating to such item. The Chairman of the Committee may co-opt technical experts as relevant to any issue or class of issues under its consideration.

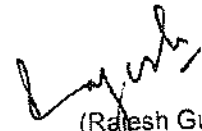
17. Functions of the Standing Committee: The Standing Committee shall meet as often as necessary, but not less than once in six months. The Committee

- a. shall oversee the implementation of this order and issues arising therefrom, and make recommendations to Nodal Ministries and procuring entities.
- b. shall annually assess and periodically monitor compliance with this Order
- c. shall identify Nodal Ministries and the allocation of items among them for issue of notifications on minimum local content
- d. may require furnishing of details or returns regarding compliance with this Order and related matters
- e. may, during the annual review or otherwise, assess issues, if any, where it is felt that the manner of implementation of the order results in any restrictive practices, cartelization or increase in public expenditure and suggest remedial measures
- f. may examine cases covered by paragraph 13 above relating to manufacture under license/ technology transfer agreements with a view to satisfying itself that adequate mechanisms exist for enforcement of such agreements and for attaining the underlying objective of progressive indigenization
- g. may consider any other issue relating to this Order which may arise.

18. Removal of difficulties: Ministries /Departments and the Boards of Directors of Government companies may issue such clarifications and instructions as may be necessary for the removal of any difficulties arising in the implementation of this Order.

19. Ministries having existing policies: Where any Ministry or Department has its own policy for preference to local content approved by the Cabinet after 1st January 2015, such policies will prevail over the provisions of this Order. All other existing orders on preference to local content shall be reviewed by the Nodal Ministries and revised as needed to conform to this Order, within two months of the issue of this Order.

20. Transitional provision: This Order shall not apply to any tender or procurement for which notice inviting tender or other form of procurement solicitation has been issued before the issue of this Order.



(Rajesh Gupta)

Director

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COMMERCIAL OFFER (UN-PRICED)

(Compliance for submitting bids for listed items below:

To be submitted as a part of techno-commercial bid)

FIRE FIGHTING SYSTEMS FOR SHED-4 & ADMIN BUILDING

SL. NO.	DESCRIPTION	QTY	APPLICABLE GST RATES	HSN	QUOTED	
					YES	NO
1	Design and Supply of materials for Fire Fighting Systems for Shed-4 & Admin Building at Rohtak plant as per (Annexure-I, II, III)	01 SET				
2	Installation, Erection, Commissioning & Testing of the above systems as per tender	01 LUMPSUM				

BILL OF QUANTITIES														
(Price Break up to be submitted strictly in Price Bid/Cover-II only)														
SL NO	ANNEXURE	DSR 2019	Description	HSN Code	DSR 2019	Sub-description	Unit	Qty.	Basic Unit Price in Rs.	SGST per unit in Rs.	CSGT per unit in Rs.	IGST per unit in Rs.	Total Amount incl GST per unit in Rs.	Total Amount incl of GST
			PIPES AND ACCESSORIES											
ITEM NO 1	FIRE FIGHTING WET RISER HYDRANT & SPRINKLE SYSTEM [ANNEXURE-I]	6	Supply of 'C' class heavy duty MS Pipe conforming to IS 1239/3589 i/c fittings like elbows, tees, flanges, tapers, nuts bolts, gaskets etc. in ground including welding, excavation & providing cement concrete blocks as supports, anticorrosive treatment with coaltar/asphalt tape as per IS 10221, refilling the trench etc. of following sizes complete as required. (MAKE: JINDAL (HISSAR), TATA, PRAKASHI)	7306	6.2	150 mm. Dia	Metre	78						
ITEM NO 2		7	Supply of 'C' class heavy duty MS pipe conforming to IS 3589/IS 1239 including Welding, fittings like elbows, tees, flanges, tapers, nuts bolts, gaskets etc. and fixing the pipe on the wall/ceiling with suitable clamp/support frame and painting with two or more coats of synthetic enamel paint of required shade complete as required : (MAKE: JINDAL (HISSAR), TATA, PRAKASHI)	7306	7.1	25 mm dia	RM	210						
ITEM NO 3					7.2	32 mm dia	RM	294						
ITEM NO 4					7.3	40 mm dia	RM	246						
ITEM NO 5					7.4	50 mm dia	RM	12						
ITEM NO 6					7.5	65 mm dia	RM	12						
ITEM NO 7					7.6	80 mm dia	RM	24						
ITEM NO 8					7.7	100 mm dia	RM	48						
ITEM NO 9					7.8	150 mm dia	RM	126						
ITEM NO 10		9	Supply Single headed external Internal hydrant valve with 1 No. 63 mm dia instantaneous FM Gunmetal/Stainless Steel coupling and cast iron wheel, ISI marked, conforming to IS 5290 (type A) with blank Gunmetal/Stainless Steel cap and chain as required : (MAKE: SAVEMAX, NEWAGE, OMEX)	8481	9.2	Single headed Stainless steel	RM	6						
ITEM NO 11		10	Supply single headed external yard hydrant valve with 1 No. 63 mm dia instantaneous FM Gunmetal/Stainless Steel coupling and cast iron wheel, ISI marked, conforming to IS 5290 (type A) with blank Gunmetal/Stainless Steel cap and chain as required : (MAKE: SAVEMAX, NEWAGE, OMEX)	8481	10.2	Single headed Stainless steel	RM	4						
ITEM NO 12		11	Supply of butterfly valve of PN 1.6 rating with bronze/gunmetal seat duly ISI marked complete with nuts, bolts, washers, gaskets conforming to IS 13095 of following sizes as required :	8481	11.4	80 mm dia	Set	10						
ITEM NO 13					11.5	100 mm dia	Set	2						
ITEM NO 14					11.6	150 mm dia	Set	3						
ITEM NO 15		13	Supply of orifice plate made out of 6 mm thick stainless steel (Grade 304) with orifice of required size to be fitted between flange & landing valve of external and internal hydrants to reduce pressure at the outlet to the level of 3.5 kg/cm2 complete as required. (MAKE: SAVEMAX, NEWAGE, OMEX)	7307			Each	6						
ITEM NO 16		16	Supply 63 mm dia, 15 m long RRL hose pipe with 63 mm dia male and female couplings duly bound with GI wire, rivets etc. conforming to IS 636 (type-A) as required:(MAKE: SAVEMAX/OMEX/NEWAGE)	5909	16.2	Stainless Steel (Grade 304)	Set	20						
ITEM NO 17		17	Supply first-aid Hose Reel with MS construction spray painted in post office red, conforming to IS 884 complete with the following as required. 20 mm nominal internal dia water hose thermoplastic (Textile reinforced) type -2 as per IS: 12585 20 mm nominal internal dia gun metal globe valve & nozzle. Drum and brackets for fixing the equipments on wall. Connections from riser with 25 mm dia stop gun metal valve & M.S. Pipe and socket.(MAKE: SAVEMAX/OMEX/ NEWAGE)	8424	17.1	30 m	Set	4						
ITEM NO 18		21	Supply 15mm dia quartzoid bulb type sprinklers of rating 68 degree centigrade with required accessories :	8424	21.1	Pendent Sprinkler	Each	40						
ITEM NO 19					21.2	Upright Sprinkler	Each	220						
ITEM NO 20		24	Supply installation control valve of cast iron body, brass/bronze working parts comprising of water motor alarm, bronze seat clapper, clapper arm and hydraulically driven mechanical gong bell to sound continuous alarm when the wet riser/sprinkler system activates, pressure gauges, emergency releases, strainer, pressure switch, cock valve complete with drain valve and bypass, test control box, ball valves, MS pipe of required size, flanges, orifice plate, gasket etc of following sizes as required :	8481	24.3	150mm dia	Set	1						
ITEM NO 21		25	Supply sprinkler flexible pipe (UL Listed) of stainless steel complete with 15 NPT on reducer thread with maximum working pressure of 175 PSI test pressure of 875 PSI (Burst) with branch line (Inlet) 25mm NPT male thread to sprinkler head (Outlet) 15mm NPT female thread with reducer, nipple, 2 side brackets, centre bracket, stockbar of following sizes complete as required.	8307	25.3	1200mm	Set	40						
ITEM NO 22		29	Supply adjustable rosette plate for 15mm dia in white finish UL Listed or FM approved complete as required.	7907			Each	40						
ITEM NO 23		1	Supply gun metal ball valve suitable for pressure 15 Kg/Cm2 and conforming to IS: 778 of the following size including providing necessary union / flange.	8481	a)	25 mm dia	NOS	4						
ITEM NO 24		5	Supply weather proof cabinet of size not less than 0.75 x 0.6 x 0.25 mtr made out of M.S. sheet not less than 1.5 mm thick having central opening and 4 mm thick glazed glass doors (Two nos.) suitably marked on the outside with the letters "FIRE HOSE" including necessary locking arrangement and painting in red colour suitable to accommodate, external yard hydrant valve, 2 nos 15 mtr long Hose pipe, branch pipe, nozzle and fire man's axe. It shall be mounted on boundary wall complete as required.	7310			NOS	6						
			DSR items											
ITEM NO 25	INTELLIGENT ADDRESSABLE FIRE DETECTION & ALARM [ANNEXURE-II]	17.2.4	Supply intelligent analog addressable photothermal detector complete with mounting base complete as required.	8531			Each	12						
ITEM NO 26		17.2.14	Supply addressable manual call point complete as required.	8531			Each	4						
ITEM NO 27		17.2.15	Supply addressable horn cum strobe complete as required.	8531			Each	4						
ITEM NO 28		17.5.2	Supply 2x1.5 sqmm Fire alarm armoured cable, 600/1000V rated with annealed copper conductor having XLPE insulation, steel wire armouring & FRLS outer sheath complete as required.	8544			Metre	800						
			Emergency Exit Light											
ITEM NO 29	EMERGENCY EXIT SIGNAGE [ANNEXURE-III]		Supply wall mounting / ceiling hung type, self contained Rechargeable LED Emergency Egress Routes maintained type Signage/ Escape Route/ Directional Light with LED lamp having multi LED illumination on normal as well as mains failure on self contained battery source complete with G.I. chain, brackets, hardwares etc. as required for fixing. ON resumption of mains supply, the lamp switches back to the mains automatically and simultaneously starts recharging the battery to the required level. The Signage Fitting shall be fabricated from 22SWG C.R.C.A-steel, Epoxy polyester powder coating, White in colour, 2 m.m. thick milky acrylic with vinyl pasted on it, complete with rechargeable Ni-Cd / Ni-Mh battery for 3 Hours backup, 16 to 24 hours recharging period with constant current charger. Input - 230V, 50 HZ, confirms to IS: 10322. (Ref PROLITE Autoglo Model -PEL LED Wall Mounted A.D. or equivalent).	9405			Each	4						
ITEM NO 30			Supply wall mounting / ceiling hung type, self contained Rechargeable LED Emergency Egress Routes maintained type Signage/ Escape Route/ Directional Light with LED lamp having multi LED illumination on normal as well as mains failure on self contained battery source complete with G.I. chain, brackets, hardwares etc. as required for fixing. ON resumption of mains supply, the lamp switches back to the mains automatically and simultaneously starts recharging the battery to the required level. The Signage Fitting shall be fabricated from 22SWG C.R.C.A-steel, Epoxy polyester powder coating, White in colour, 2 m.m. thick milky acrylic with vinyl pasted on it, complete with rechargeable Ni-Cd / Ni-Mh battery for 3 Hours backup, 16 to 24 hours recharging period with constant current charger. Input - 230V, 50 HZ, confirms to IS: 10322. (Ref PROLITE Autoglo Model -PEL LED Wall Mounted A.D. or equivalent).	9405			Each	2						
ITEM NO 31	SERVICE		Installation, Commissioning & Testing of the complete firefighting Systems				Lum	1						
													TOTAL AMOUNT INCLUSIVE OF GST	

PRICE BID FORMAT

FIRE FIGHTING SYSTEMS FOR SHED-4 & ADMIN BUILDING

SL. NO.	DESCRIPTION	QTY	UOM	BASIC UNIT PRICE (in Rs)	GST (in Rs)	TOTAL AMOUNT (in Rs)
1	Design and Supply of materials for Fire Fighting Systems for Shed-4 & Admin Building at Rohtak plant as per (Annexure-I, II, III)	01	SET	To be quoted & submitted in Price Cover	To be quoted & submitted in Price Cover	To be quoted & submitted in Price Cover
2	Installation, Erection, Commissioning & Testing of the above systems as per tender	01	LUMPSUM	To be quoted & submitted in Price Cover	To be quoted & submitted in Price Cover	To be quoted & submitted in Price Cover

Note: Price Break up of the Design & Supply of materials as per Annexure-I, II & III in the price bid cover as per attached format

FORM OF BANK GUARANTEE IN LIEU OF SECURITY DEPOSIT

1. This deed of guarantee executed onday of by
.....

(Name and Address of the Bank)

the Bank hereinafter called Bank (which term shall mean and include its successors and assigns wherever the context so admits) in favour of M/s. MISHRA DHATU NIGAM LIMITED., a government of India Enterprise incorporated and registered as a company under the Companies Act, 1956, having its registered office at P.O. Kanchanbagh, Hyderabad – 500 058, state of A.P. India, herein after referred to as the "purchaser" (which terms shall mean and include its successors in office and assigns).

2. In consideration of M/s. Mishra Dhatu Nigam Limited (Purchaser) having agreed to exempt hereinafter called the said Contractor(s) (which term shall mean and include its successors assigns and legal representatives) from the demand under the terms and conditions of Purchase / Work Order No. dated for(hereinafter called the said agreement) of Earnest Money / Security Deposit/ Defect Liability deposit for the due fulfillment by the said Contractor(s) of the terms and conditions contained in the said agreement on production of a bank guarantee for Rs.....(Rupees..... only), we (name of the bank, address) (hereinafter referred to as "The Bank") at the request of Contractor(s) do hereby undertake to pay Purchaser an amount not exceeding Rs against any losses or damage caused to or suffered or would be caused to or suffered Purchaser by reason of any breach by the said Contractor(s) of any of the terms and conditions contained in the said agreement.
3. We (Bank) do hereby unconditionally and irrevocably agree and undertake to pay to Purchaser the amounts due and payable under this Guarantee without any demur, merely on a demand from Purchaser stating that the amount claimed is due by way of loss or damage caused to or would be caused to on suffered by Purchaser by reason of breach by the said agreement or by reason of the contractor(s) failure to perform the said agreement. Any such demand made on the Bank shall be conclusive as regards the amount due and payable by the Bank under this Guarantee. However, our liability under this Guarantee shall be restricted to an amount not exceeding Rs
.....
4. We undertake to pay Purchaser and money so demanded notwithstanding any dispute or disputes by the contractor(s) / supplier(s) in any suit or proceedings pending before any court of tribunal relating thereto, our liability under this present being absolute and unequivocal. The payment so made by us under this Guarantee shall be valid discharge of our liability for payment thereunder and the contractor(s) shall have not claim against us for making such payment.

5. We (Bank) further agree that the guarantee herein contained shall remain in full force and affect during the period that would be taken for the performance of the said Agreement and that it shall continue to be enforceable till all the dues of Purchase under or by virtue of the said Agreement have been fully paid and its claims satisfied or discharges or till Purchaser certifies that the terms and conditions of the said Agreement have been fully and properly carried out by the said contractor(s) and accordingly discharges this guarantee. Provided that if Purchaser together with the Contractor seeks an extension of terms of the Guarantee, such extension shall be granted by the Bank and the guarantee shall be in full force till the expiry of such extended period.
6. We (Bank) further agree with Purchaser that Purchaser shall have the fullest liberty without our consent and without affecting in any manner our obligations hereunder to vary any of the terms and conditions of the said agreement or to extend time of performance by the said contractor(s) from time to time or to postpone for any time or from time to time any of the powers exercisable by purchaser against the said Contractor(s) and to forebear or enforce any of the terms and conditions relating to the said agreement and we shall not be relieved from our liability by reason of any such variation of extension being granted to the said contractor(s) or for any forbearance, act or omission on the part of Purchaser or any indulgence by Purchaser to the said contractor(s) or by any such matter or thing whatsoever which under the law relating to sureties would but for this provisions, have effect of so relieving us.
7. It shall not be necessary for Purchaser to proceed against the contractor before proceeding against the Bank and the guarantee herein contained shall be enforceable against the Bank notwithstanding any security which Purchaser may have obtained or obtains from the contractor.
8. This guarantee shall not be discharged due to the change in the constitution of the Bank or the contractor(s).
9. We (Bank) lastly undertake not to revoke this guarantee during its currency except with the previous consent of Purchase in writing.
10. Our liability is limited to a sum not exceeding Rs unless a claim is made on us in writing on or before (3 months beyond the date of delivery / completion as specified in the contract) we shall be discharged from liability under this guarantee.

In witness whereof these presents are executed at on the date, month and year first herein above written.

FOR AND ON BEHALF OF THE BANK WITHIN NAMED

BANK GUARANTEE TOWARDS PERFORMANCE BANK GUARANTEE

This Deed of Guarantee executed on..... Day of..... by..... Bank. hereinafter called the "Guarantor" (which term shall mean and include its successors, assigns and legal representatives, where the context so admits) in favour of M/s. MISHRA DHATU NIGAM LIMITED, a Government of India Enterprise incorporated and registered as a Company under the Companies Act, 1956, having its registered office at P.O. Kanchanbagh, Hyderabad - 500 058, State of A.P. INDIA, hereinafter referred to as the "Purchaser", (which term shall mean and include its successors and assigns).

WHEREAS the Purchase Order No: MDN/PUR/..... (*Indicate both the order Numbers*) Dated..... has been entered into between the Purchaser and M/s..... a company incorporated and registered under the Laws of by which the company is permitted to manufacture and sell certain products, and having its registered office situated at hereinafter referred to as the SUPPLIER (which term shall mean and include its successors, assigns and legal representatives) for the supply of (*Indicate the list of Items which are replaced/repared*) as fully described in the aforesaid Purchase Order.

AND WHEREAS it has been agreed under the terms and conditions of the aforesaid Purchase Order that the Purchaser shall make balance payment to the Supplier of(amount) in words (.....) representing **10%** of the total value of the repaired and replaced parts in the Purchase Orders on final acceptance of the and on furnishing a Guarantee from a Bank acceptable to the Purchaser by the Supplier towards satisfactory performance of the supplied.

AND WHEREAS the Guarantor has, as per the terms and conditions of the aforesaid Purchase Order, agreed to stand guarantee for the amount of ten per cent of the balance payment in favour of the Supplier and the Guarantor is acceptable to the Purchaser.

NOW THIS DEED witnesses that, in pursuance of the terms and conditions of the aforesaid Purchase Order and in consideration of the payment of representing **10%** of the total value of the repaired and replaced parts in the Purchase Orders agreed to be made to the Supplier by the Purchaser, the Guarantor do hereby agree and undertake to indemnify the Purchaser and keep the purchaser indemnified to the extent of a sum not exceeding the said sum of (Amount) (In Words) against any damage or loss that may be suffered by the Purchaser by reason of non-fulfillment of or breach any of the terms and conditions of the Purchase Order by the supplier, and the guarantor hereby undertaken to pay on demand and without any demur to the purchaser any sum unconditionally and irrevocably not exceeding the sum of (Amount) (In Words) as may be demanded by the Purchaser's General Manager (Commercial) or Representative of the purchaser as the damages or loss that the purchaser may have suffered by reasons of non-fulfillment of any of the terms and

conditions of the Purchase Order by the Supplier, and Guarantor hereby covenants with the Purchaser as follows:

1. That the decision of the General Manager (Commercial) or Representative of the Purchaser as to whether the said installation under the Purchase Order gives satisfactory performance or not and as to the amount of damages suffered by the Purchaser on account of the unsatisfactory performance of the said installation under the Purchase Order shall be conclusive, final and binding on the Bank.
2. That the Guarantee herein contained shall remain in full force and effect till the Purchaser certifies in writing that the terms and conditions of the said Purchase Order have been fully and properly carried out by the said Supplier and accordingly discharges the Guarantee. Unless a demand for claim under this Guarantee is made on the Bank in writing on or before, the Bank shall be discharged from all liabilities under this Guarantee thereafter, provided that if the Purchaser, together with the Supplier, seeks an extension of the term of the Guarantee, such extension shall be granted by the Bank and the Guarantee shall be in full force and effect till the expiry of such extended period.
3. That the Purchaser shall have the fullest liberty, without affecting in any way the liability of the Bank under this Guarantee or Indemnity, from time to time to vary any of the terms and conditions of the said Purchase Order or to extend its performance by the said Supplier as provided or to postpone, for any time and from time to time, any of the powers exercisable by it against the said Supplier and either to enforce or forbear from enforcing any of the terms and conditions governing the said Purchase Order, and the said Bank shall not be released from its liability under these presents by any exercise by the Purchaser of the liberty with reference to matters aforesaid or by reason of time being given to the said Supplier or any other forbearance, act or omission on the part of the Purchaser or any indulgence by the Purchaser to the said Supplier or of any other matter or thing whatsoever which, under the law relating to sureties, would, but for this provision, have the effect of so releasing the Bank from its liability.
4. It shall not be necessary for the Purchaser to proceed against the Supplier before proceeding against the Bank and the Guarantee herein contained shall be enforceable against the Bank notwithstanding any security which the Purchaser may have obtained or obtain from the Supplier.
5. The Bank lastly undertakes not to revoke this Guarantee during its currency, except with the previous consent of the Purchaser in writing, and agrees that any change in the constitution of the said Supplier or the said Bank shall not discharge the Bank's liability hereunder.

In witness whereof these presents are executed at the date, month and year, first herein above written.

FOR & ON BEHALF OF THE BANK WITHIN NAMED